

# CO<sub>2</sub> Storage Modeling and Rock Physics Agent Development

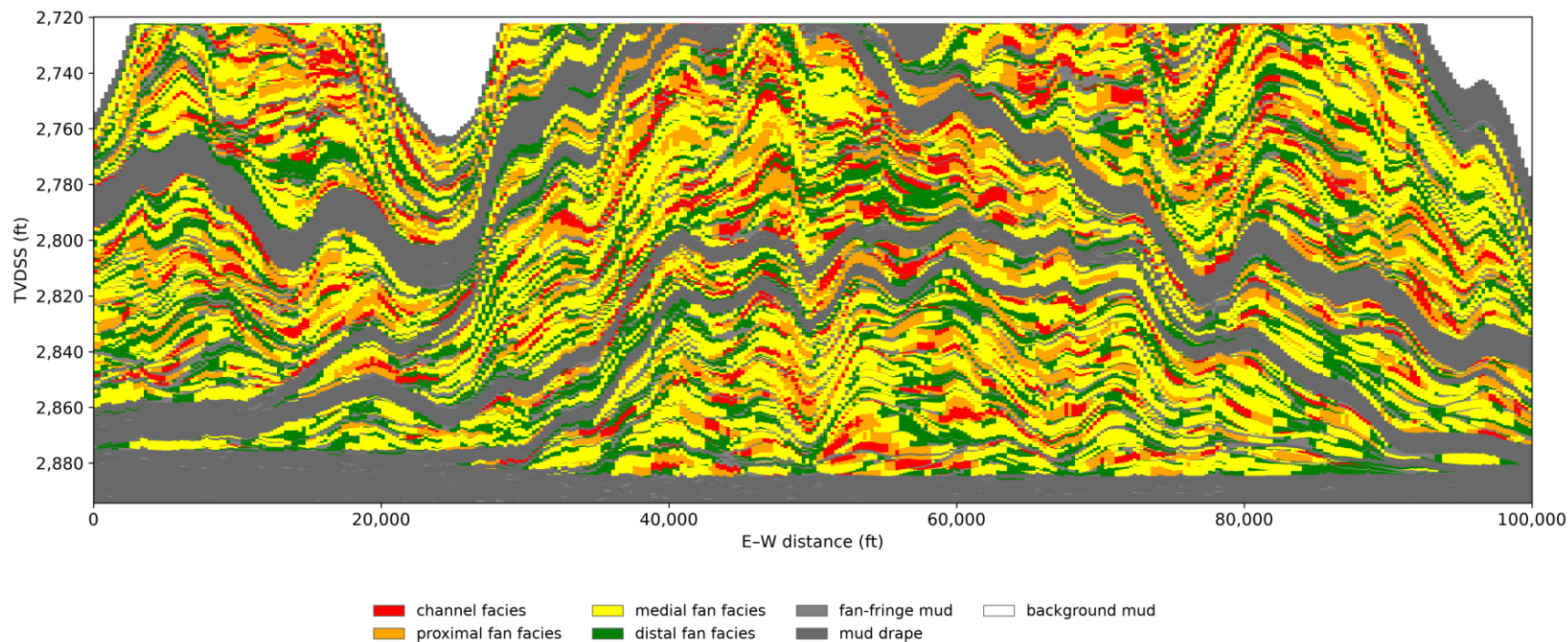
Buyu Deng

*September 25, 2026*

*SCERF Seminar*

# Geological Model Section View

**Geological Model · Section View**

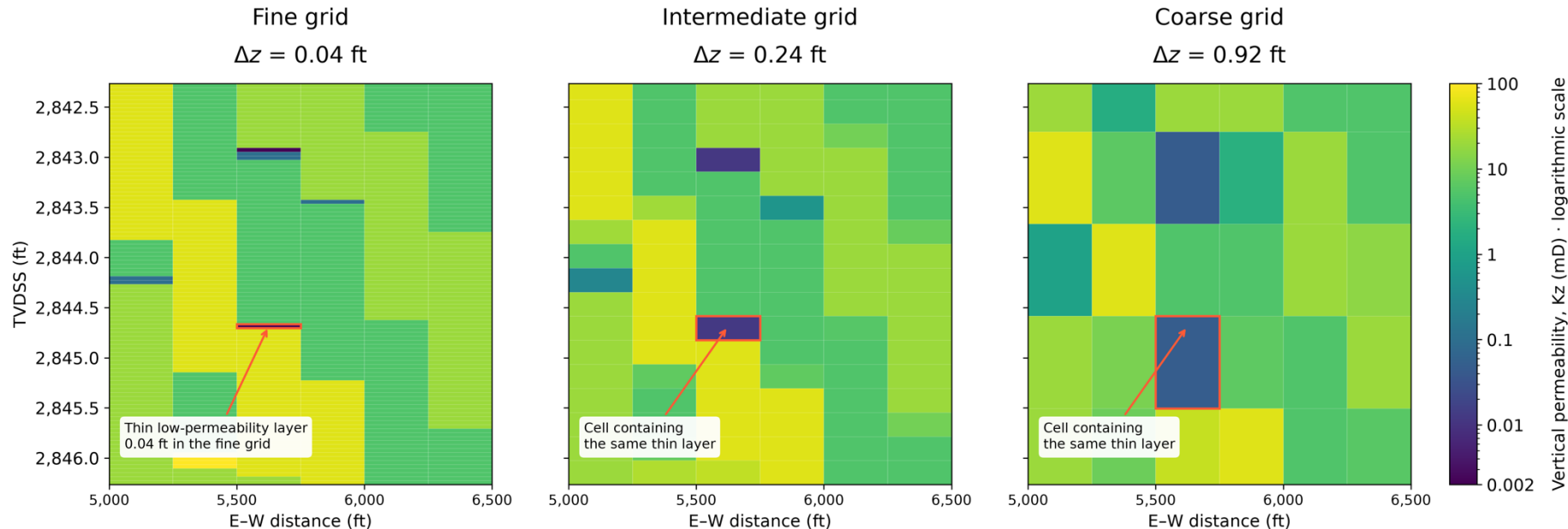


# Seven Facies Define a Sand-to-Mud Depositional Continuum

| Code | Facies                      | Lithology                          | Load-bearing interpretation |
|------|-----------------------------|------------------------------------|-----------------------------|
| 1    | Channel                     | Relatively clean sandstone         | Sand-grain supported        |
| 2    | Proximal fan                | Slightly muddy sandstone           | Sand-grain supported        |
| 3    | Medial fan                  | Shaly sandstone                    | Sand-grain supported        |
| 4    | Distal fan                  | Mud-rich shaly sandstone           | Sand-grain supported        |
| 0    | Fan-fringe                  | Muddy siltstone / sand–mud mixture | Mud-matrix supported        |
| 9    | Mud drape                   | Mudstone                           | Mud-matrix dominated        |
| 99   | late-stage mudstone deposit | Mudstone                           | Mud-matrix dominated        |

# Vertical Resolution and Simulation Cost

Coarser grids reduce computational cost but change thin-layer representation.



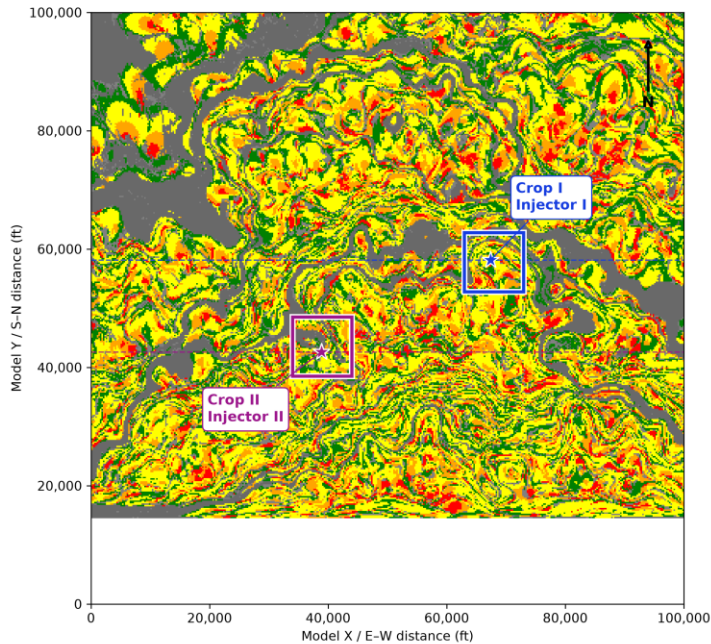
**Objective: Find the coarsest vertical grid that preserves key CO<sub>2</sub> flow predictions within an acceptable tolerance.**

# Two Injectors and Two Crops

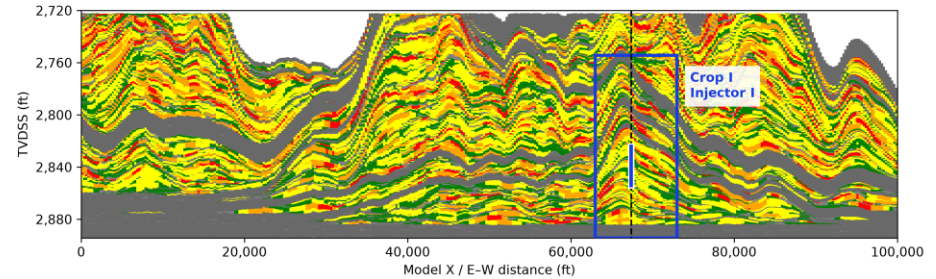
## Geological Model • Two Injectors and Two Crops

Full model: 100,000 × 100,000 × 174.36 ft | Each crop: 10,000 × 10,000 × 140 ft

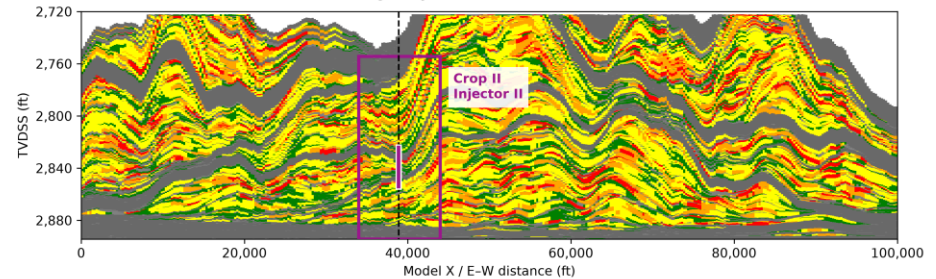
Plan view • TVDSS = 2,839.01 ft



E-W through Injector I • model Y = 58,125 ft

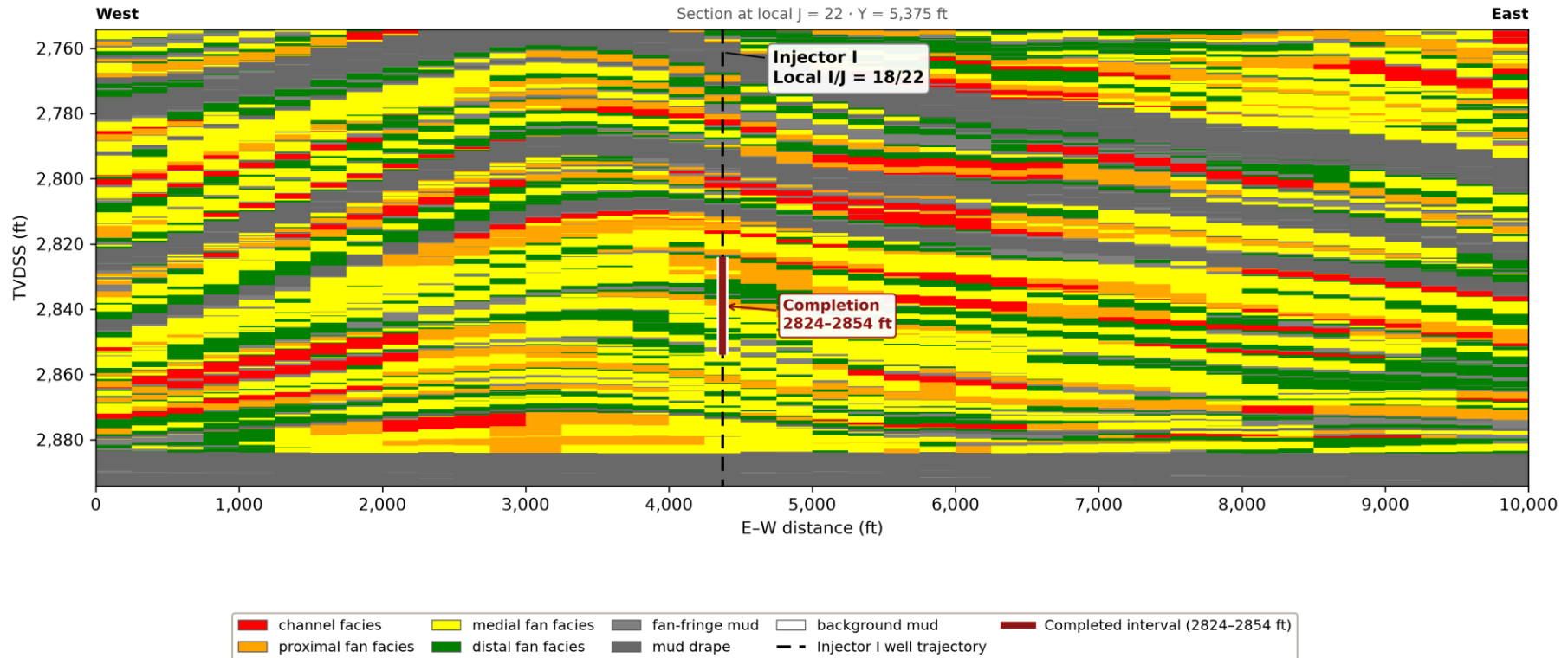


E-W through Injector II • model Y = 42,625 ft



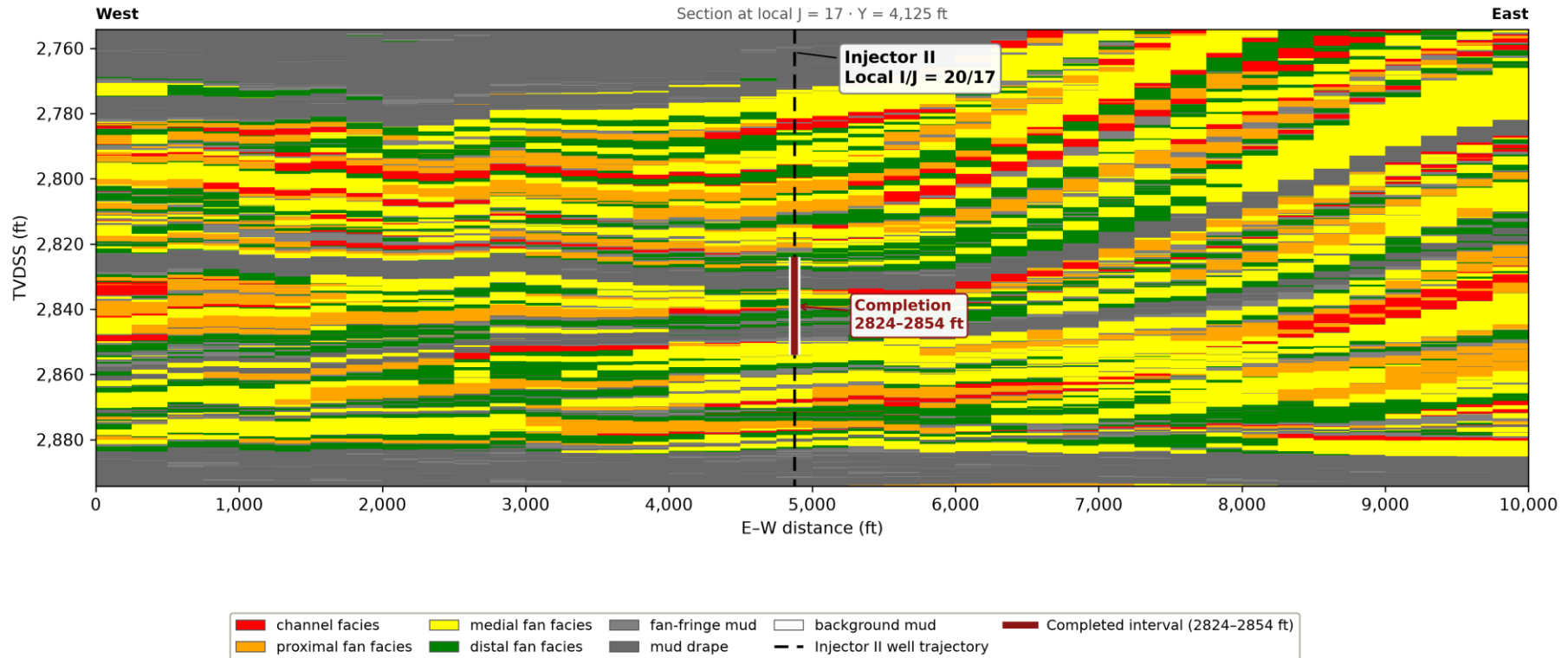
# Crop I: Geological E–W Section Through Injector

Crop I · Geological E–W Section Through Injector I



# Crop II: Geological E–W Section Through Injector

Crop II · Geological E–W Section Through Injector II



# Simulation Framework and Grid

| Category                | Parameter                | Setting                                                                                                 |
|-------------------------|--------------------------|---------------------------------------------------------------------------------------------------------|
| Simulator               | Simulation framework     | ECLIPSE 300, CO <sub>2</sub> STORE, fully implicit                                                      |
| Fluid system            | Components               | H <sub>2</sub> O, CO <sub>2</sub> , NaCl, CaCl <sub>2</sub> ; pure CO <sub>2</sub> injection into brine |
| Initial conditions      | Pressure and temperature | 105 bar at 2,824.27 ft TVDSS; 32.2°C                                                                    |
| Model dimensions        | Domain and lateral grid  | 10,000 × 10,000 × 140 ft; 40 × 40 lateral cells; $\Delta x = \Delta y = 250$ ft                         |
| Vertical discretization | Nominal layer thickness  | 0.04, 0.08, 0.12, 0.24, and 0.92 ft                                                                     |
| Permeability anisotropy | Fine-grid relationship   | $K_x = K_y$ ; $K_z/K_h = 0.1$                                                                           |
| Vertical upscaling      | Property averaging       | Arithmetic averaging of porosity and $K_x/K_y$ ; harmonic averaging of $K_z$                            |



# Facies Properties

| Category                                  | Parameter                          | Setting         |
|-------------------------------------------|------------------------------------|-----------------|
| Channel facies                            | Porosity / horizontal permeability | 0.27 / 1,000 mD |
| Proximal fan facies                       | Porosity / horizontal permeability | 0.25 / 600 mD   |
| Medial fan facies                         | Porosity / horizontal permeability | 0.22 / 200 mD   |
| Distal fan facies                         | Porosity / horizontal permeability | 0.18 / 50 mD    |
| Fan-fringe mud                            | Porosity / horizontal permeability | 0.08 / 1 mD     |
| Mud drape and late-stage mudstone deposit | Porosity / horizontal permeability | 0.03 / 0.02 mD  |

# Injection and Boundary Conditions

| Category            | Parameter                      | Setting                                                                                        |
|---------------------|--------------------------------|------------------------------------------------------------------------------------------------|
| Multiphase flow     | Relative permeability          | Same prescribed water/gas relative-permeability curves across all cases                        |
| Injector            | Well configuration             | One vertical injector per crop; target completion 70–100 ft below the model top                |
| Injection control   | Target rate and pressure limit | 0.1 Mt/year equivalent to 147,376.8 Sm <sup>3</sup> /day; maximum BHP = 599 bar                |
| Boundary conditions | Lateral / vertical boundaries  | Four lateral faces connected to a constant-pressure analytical aquifer; no-flow top and bottom |
| Simulation duration | Injection period               | 20 years                                                                                       |
| Simulation outputs  | Primary fields                 | Pressure, gas saturation (SGAS), and water saturation (SWAT)                                   |

# Crop II: Geological E–W Section Through Injector

## Year 10 · Crop I

Plan view and E–W section through the injector

★ Injector I

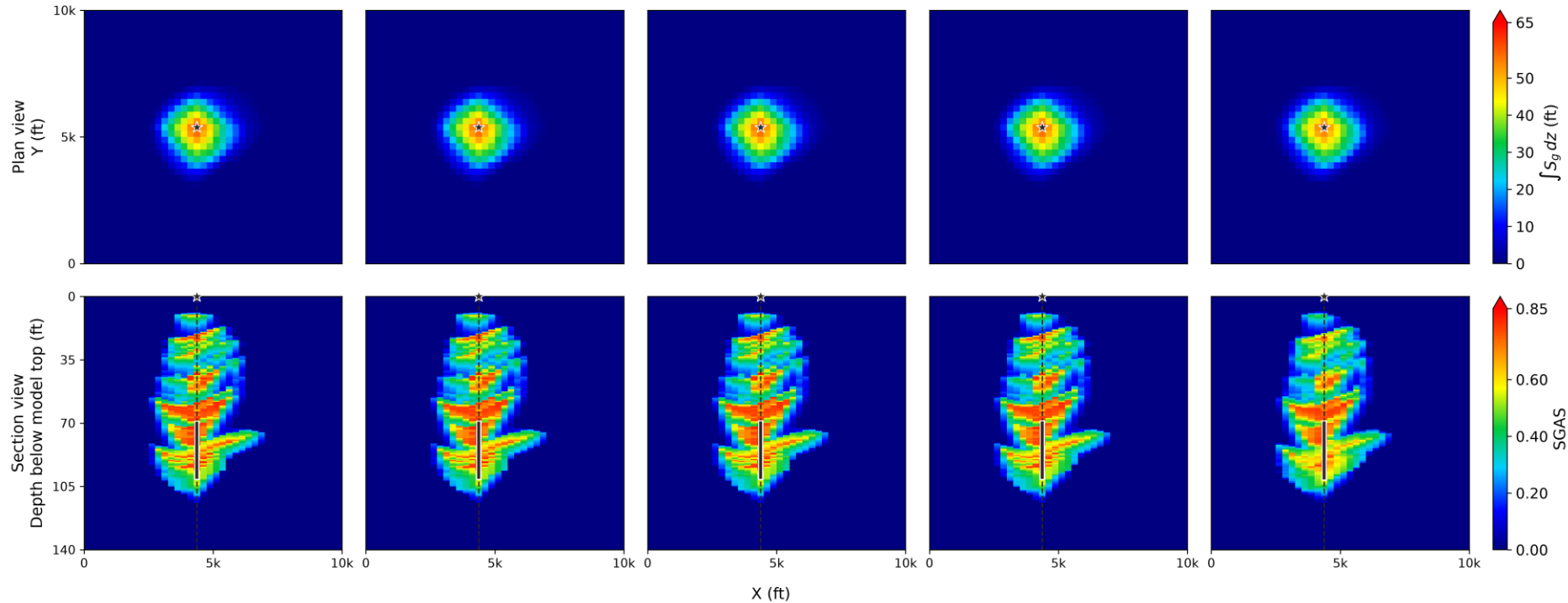
$\Delta z = 0.04$  ft

$\Delta z = 0.08$  ft

$\Delta z = 0.12$  ft

$\Delta z = 0.24$  ft

$\Delta z = 0.92$  ft



# Crop II: Geological E–W Section Through Injector

## Year 20 • Crop I

Plan view and E–W section through the injector

★ Injector I

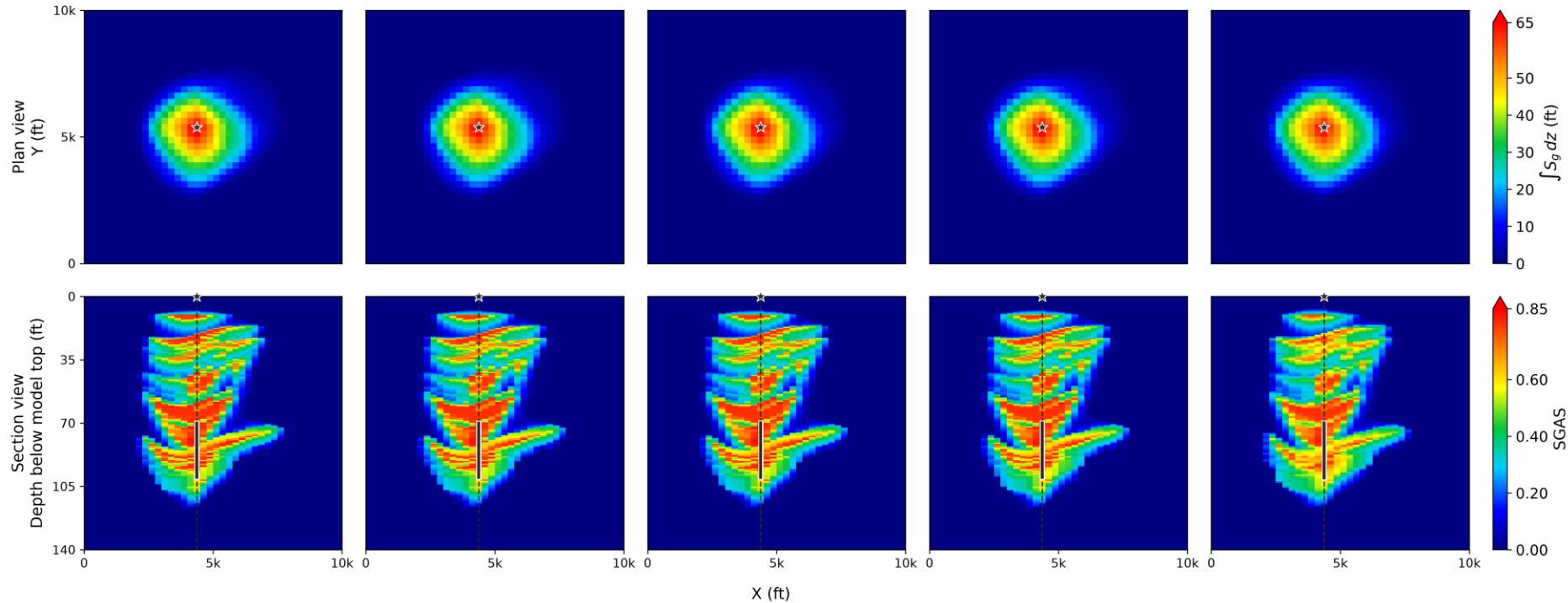
$\Delta z = 0.04$  ft

$\Delta z = 0.08$  ft

$\Delta z = 0.12$  ft

$\Delta z = 0.24$  ft

$\Delta z = 0.92$  ft



# Crop II: Geological E–W Section Through Injector

## Year 10 • Crop II

Plan view and E–W section through the injector

★ Injector II

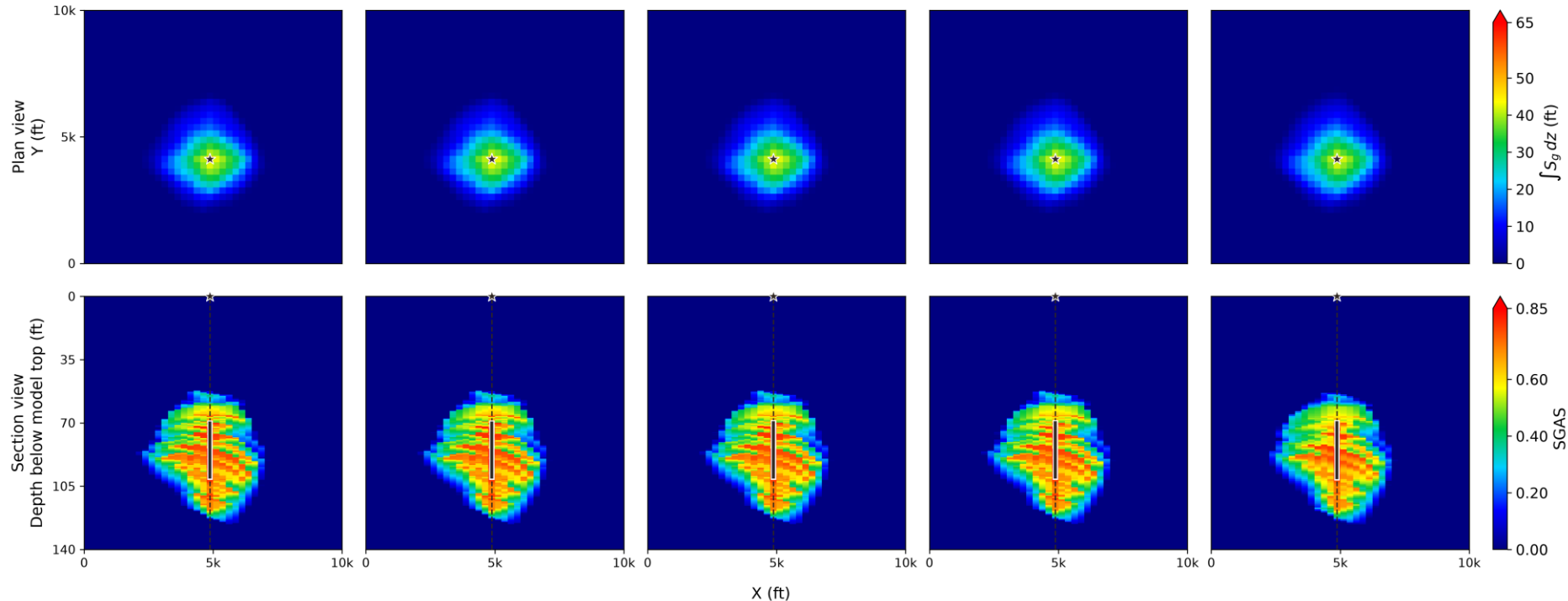
$\Delta z = 0.04$  ft

$\Delta z = 0.08$  ft

$\Delta z = 0.12$  ft

$\Delta z = 0.24$  ft

$\Delta z = 0.92$  ft



# Crop II: Geological E–W Section Through Injector

## Year 20 • Crop II

Plan view and E–W section through the injector

★ Injector II

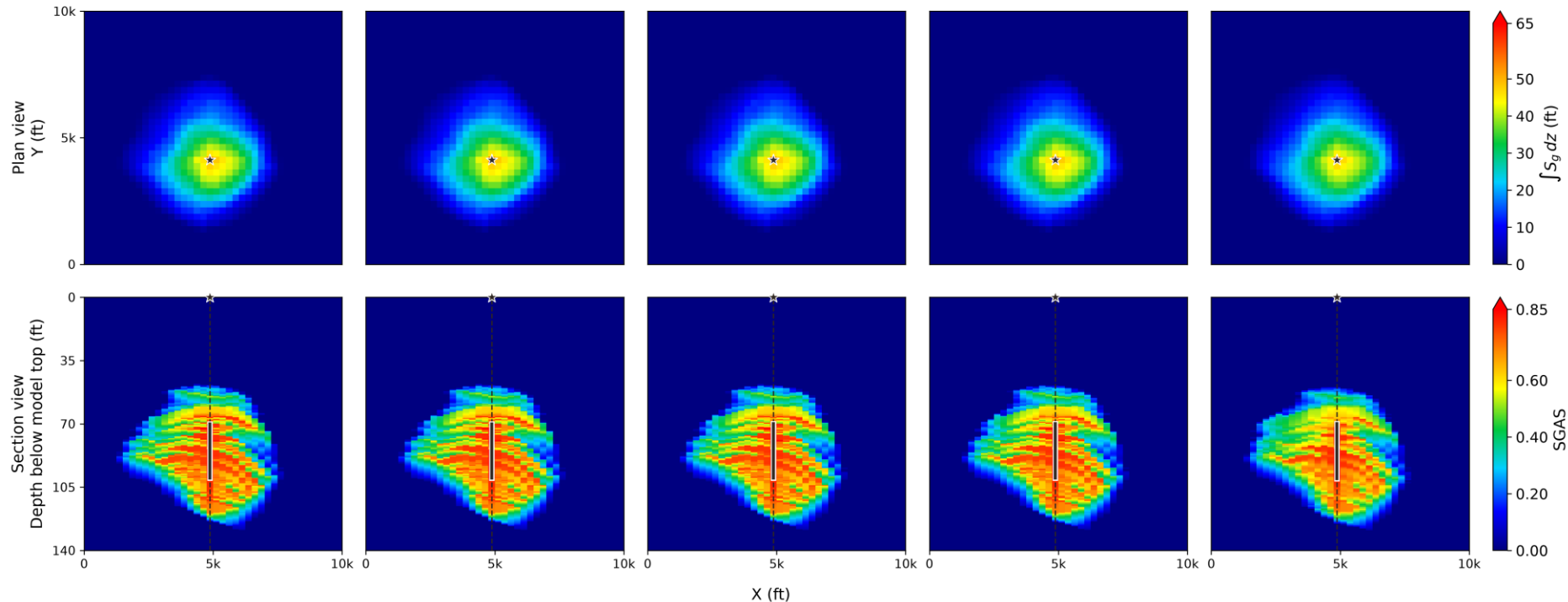
$\Delta z = 0.04$  ft

$\Delta z = 0.08$  ft

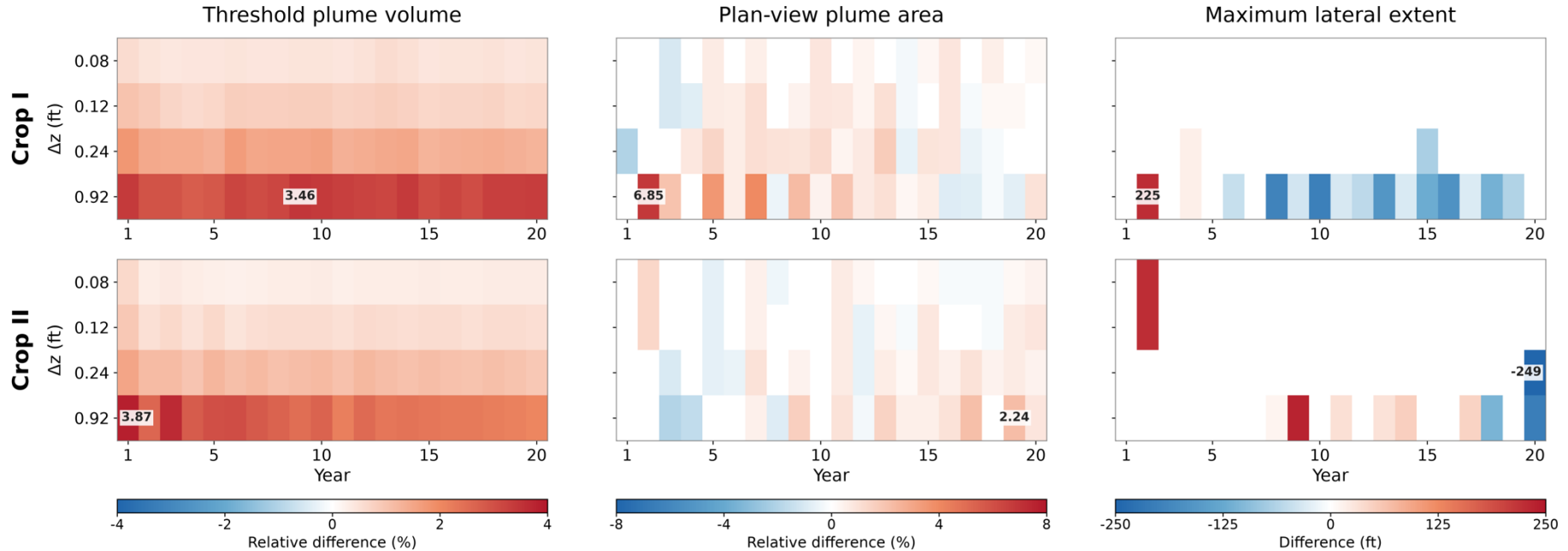
$\Delta z = 0.12$  ft

$\Delta z = 0.24$  ft

$\Delta z = 0.92$  ft



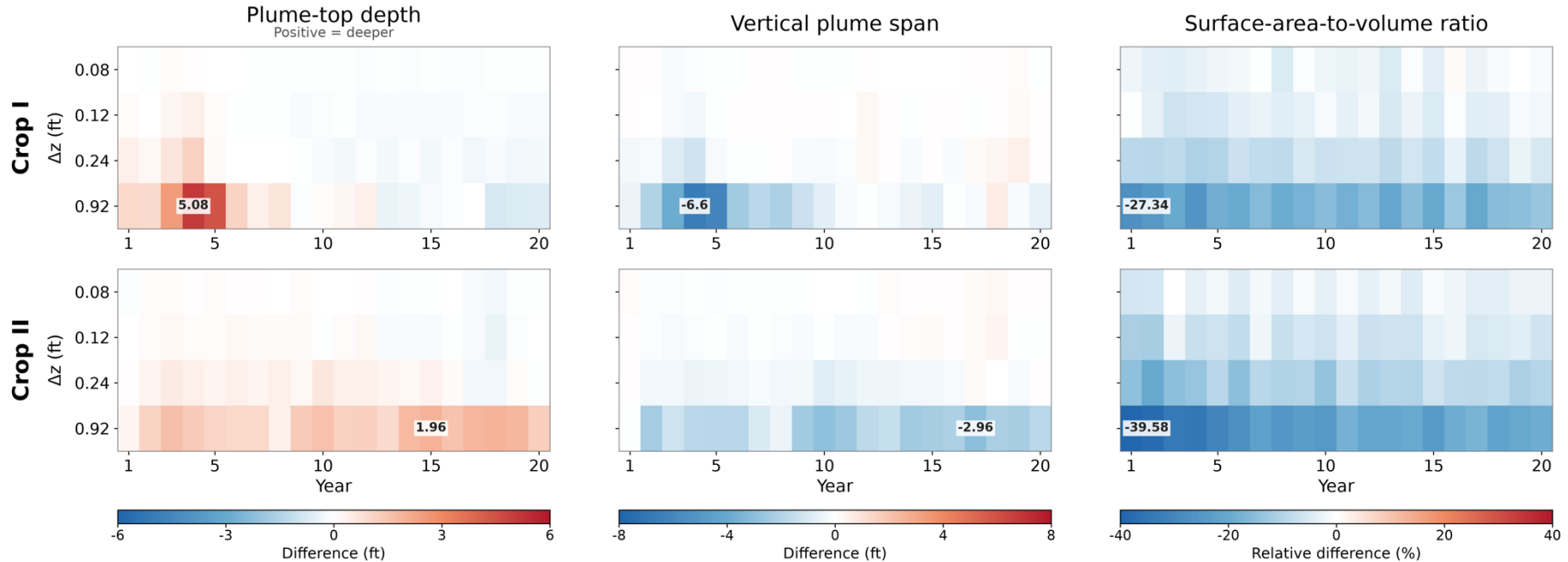
# Plume Extent: Resolution Sensitivity



Reference:  $\Delta z = 0.04$  ft. Plume:  $SGAS \geq 0.01$ . Labels mark the largest |difference| per panel.

**Volume differences stay below 4%, while lateral extent shifts by up to one 250 ft cell.**

# Plume Geometry: Resolution Sensitivity



Reference:  $\Delta z = 0.04$  ft. Plume:  $SGAS \geq 0.01$ . Labels mark the largest |difference| per panel.

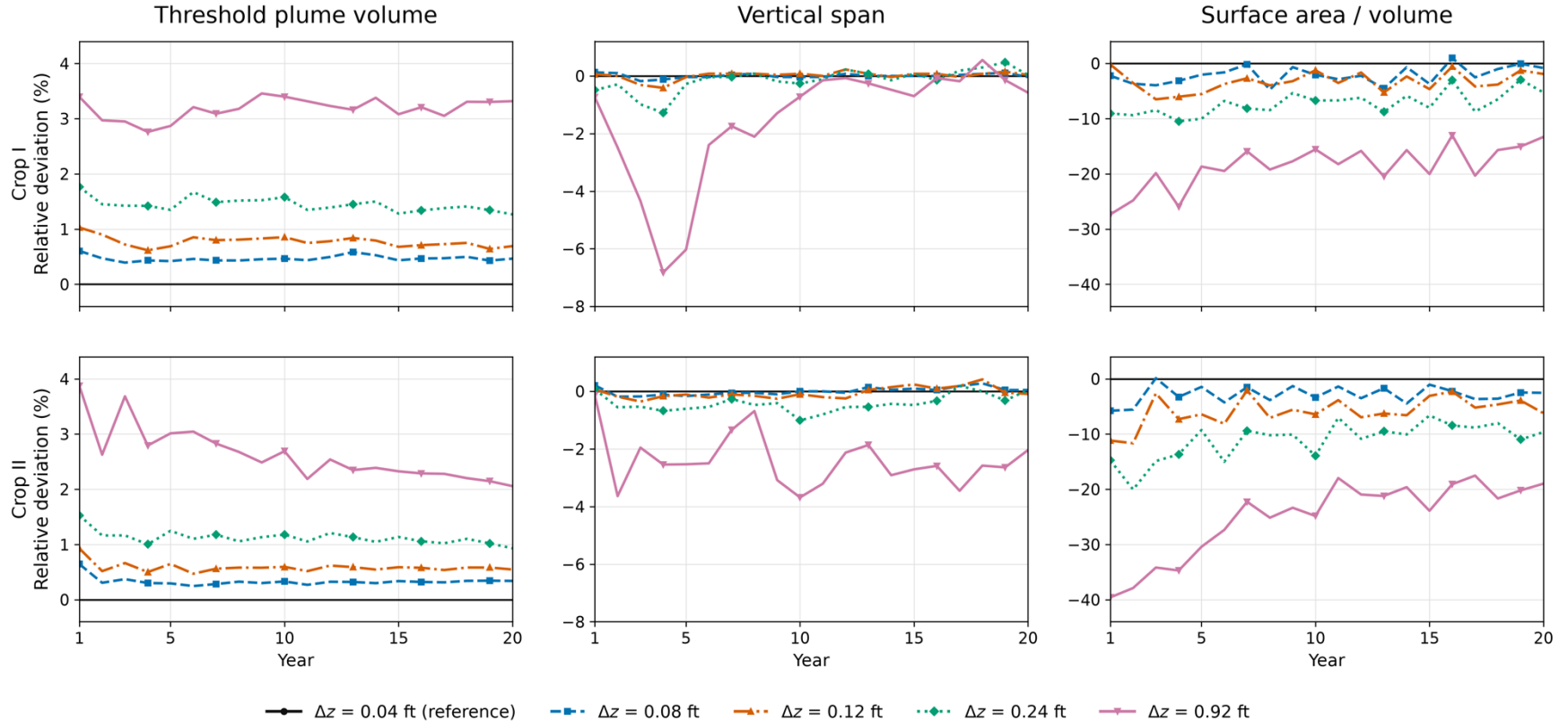
**Coarsening reduces SA/V by up to 40%, with smaller vertical shifts.**

SA/V also depends on how the grid represents plume boundaries.



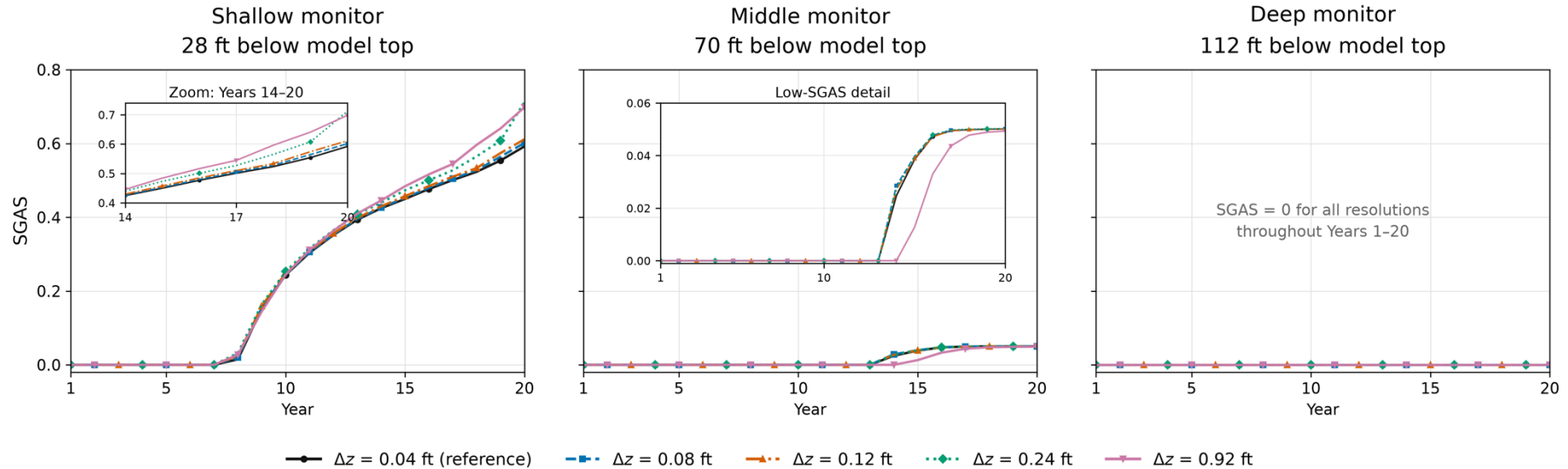
# Quantifying Resolution Sensitivity

Relative to 0.04 ft; threshold plume defined by  $SGAS \geq 0.01$ .



# Local Saturation Response: Crop I

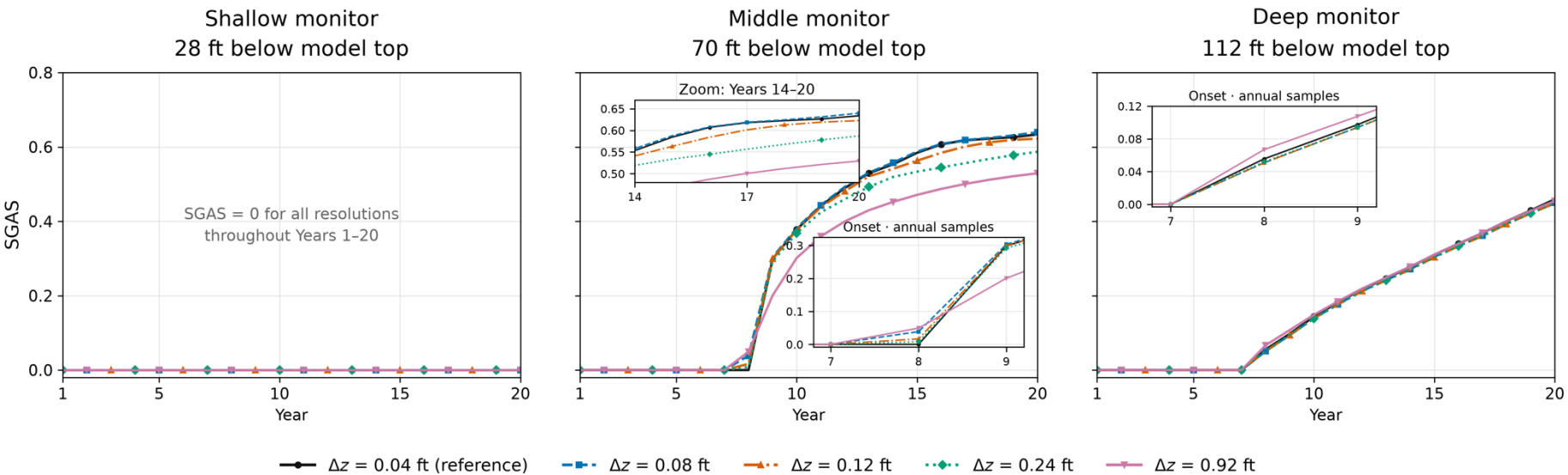
Fixed monitors 1,500 ft east of Injector I



**Shallow saturation differs; the middle monitor shows an arrival shift.**

# Local Saturation Response: Crop II

Fixed monitors 1,500 ft east of Injector II



The middle monitor is more sensitive to resolution than the deep monitor.

# Global Space-Time Metrics

Years 0–20. Threshold plume: SGAS  $\geq 0.01$ .

| Crop | $\Delta z$<br>(ft) | Time-integrated volume<br>( $10^9 \text{ ft}^3 \cdot \text{yr}$ ) | Max. lateral distance<br>(ft) | Max. vertical span<br>(ft) |
|------|--------------------|-------------------------------------------------------------------|-------------------------------|----------------------------|
| I    | 0.04               | 13.566                                                            | 3,508.92                      | 110.12                     |
| I    | 0.08               | 13.630                                                            | 3,508.92                      | 110.08                     |
| I    | 0.12               | 13.668                                                            | 3,508.92                      | 110.16                     |
| I    | 0.24               | 13.758                                                            | 3,508.92                      | 110.16                     |
| I    | 0.92               | 14.002                                                            | 3,508.92                      | 109.48                     |
| II   | 0.04               | 11.621                                                            | 3,758.32                      | 88.28                      |
| II   | 0.08               | 11.658                                                            | 3,758.32                      | 88.32                      |
| II   | 0.12               | 11.687                                                            | 3,758.32                      | 88.20                      |
| II   | 0.24               | 11.748                                                            | 3,508.92                      | 88.32                      |
| II   | 0.92               | 11.903                                                            | 3,553.17                      | 86.48                      |

# Computational Cost Across Resolutions

ECLIPSE 300. Both crops reached Year 20 with zero reported PRT errors.

| Crop | $\Delta z$<br>(ft) | Active cells<br>(million) | CPU model†     | Nodes | MPI<br>ranks | Threads/<br>rank | Elapsed<br>(h) | Recorded<br>interval |
|------|--------------------|---------------------------|----------------|-------|--------------|------------------|----------------|----------------------|
| I    | 0.04*              | 5.6                       | AMD EPYC 7742  | 1     | 100          | 1                | 24.22          | Years 3–20           |
| I    | 0.08               | 2.8                       | AMD EPYC 7742  | 1     | 50           | 1                | 7.94           | Years 0–20           |
| I    | 0.12               | 1.8672                    | AMD EPYC 7543  | 1     | 25           | 1                | 3.64           | Years 0–20           |
| I    | 0.24               | 0.9344                    | AMD EPYC 7502  | 1     | 16           | 1                | 1.62           | Years 0–20           |
| I    | 0.92               | 0.2448                    | AMD EPYC 7543  | 1     | 10           | 1                | 0.20           | Years 0–20           |
| II   | 0.04*              | 5.6                       | AMD EPYC 7742  | 1     | 100          | 1                | 25.53          | Years 3–20           |
| II   | 0.08               | 2.8                       | AMD EPYC 9384X | 1     | 50           | 1                | 3.60           | Years 0–20           |
| II   | 0.12               | 1.8672                    | AMD EPYC 9754  | 1     | 25           | 1                | 3.90           | Years 0–20           |
| II   | 0.24               | 0.9344                    | AMD EPYC 7502  | 1     | 16           | 1                | 1.42           | Years 0–20           |
| II   | 0.92               | 0.2448                    | AMD EPYC 7502  | 1     | 10           | 1                | 0.18           | Years 0–20           |

\* 0.04 ft: successful Year 3–20 restart stage only.

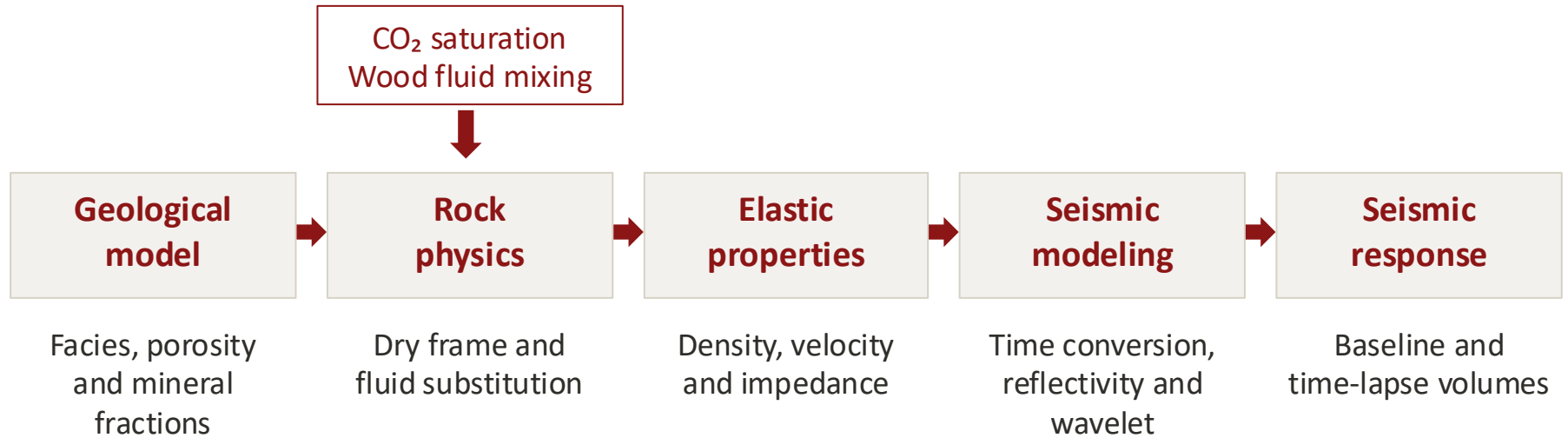
# Selected Vertical Resolution

$$\Delta z = 0.12 \text{ ft}$$

A practical starting point for subsequent modeling

# Rock Physics and Seismic Forward Modeling

Geology and simulated fluid changes become synthetic seismic responses.



Baseline: brine-saturated rock. Monitor: updated saturation at each selected year.

# Facies-Based Rock Physics Model

## Shared calculation

### Mineral mixture

Quartz + equivalent clay; VRH averaging



### Dry frame

Hertz–Mindlin + modified HS lower bound



### Saturated rock

Gassmann substitution;  $AI = \rho \times V_p$

## Facies-dependent frame parameters

| Facies group        | $\phi_c$ | n   |
|---------------------|----------|-----|
| 1–4: sand-supported | 0.40     | 8.5 |
| 0: muddy / silty    | 0.50     | 6.0 |
| 9, 99: mudrock      | 0.55     | 5.0 |

$\phi_c$ : critical porosity

n: coordination number

Fixed effective pressure: 20 MPa. Frame and mineral parameters are working priors.



# Rock Physics Parameters

Solid mineral fractions; fixed effective pressure of 20 MPa

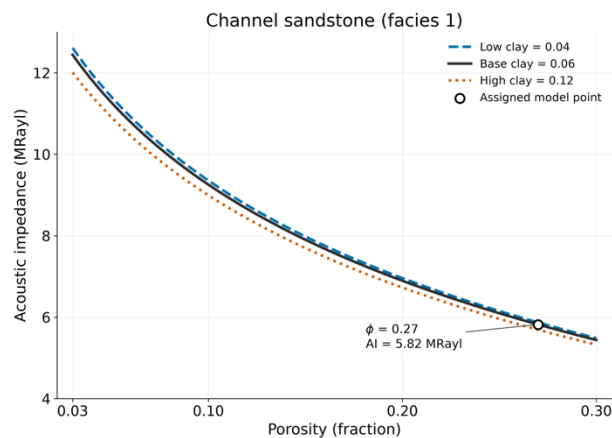
| Facies | Description            | $\phi$ | Quartz | Clay | $\phi_c$ | n   |
|--------|------------------------|--------|--------|------|----------|-----|
| 1      | Channel sandstone      | 0.27   | 0.94   | 0.06 | 0.40     | 8.5 |
| 2      | Proximal sandstone     | 0.25   | 0.90   | 0.10 | 0.40     | 8.5 |
| 3      | Medial shaly sandstone | 0.22   | 0.83   | 0.17 | 0.40     | 8.5 |
| 4      | Distal shaly sandstone | 0.18   | 0.76   | 0.24 | 0.40     | 8.5 |
| 0      | Muddy siltstone        | 0.08   | 0.65   | 0.35 | 0.50     | 6.0 |
| 9      | Mud drape              | 0.03   | 0.20   | 0.80 | 0.55     | 5.0 |
| 99     | Late-stage mudstone    | 0.03   | 0.20   | 0.80 | 0.55     | 5.0 |

$\phi$ : porosity;  $\phi_c$ : critical porosity; n: coordination number.

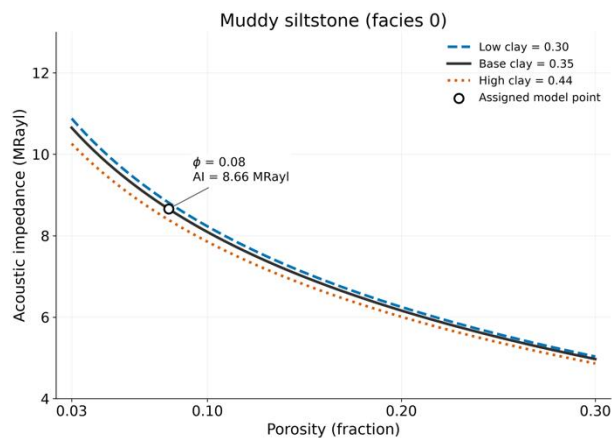
Assigned model parameters are working priors, without independent core or log calibration.

# Elastic Response to Porosity and Clay

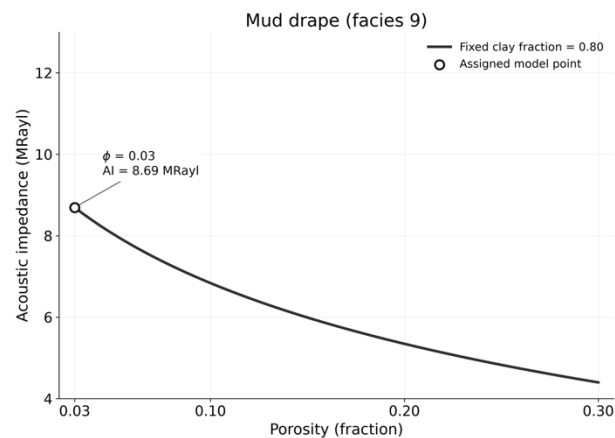
Open circles mark the parameter values assigned to the model.



$\phi = 0.27$ ; AI = 5.82 MRayl



$\phi = 0.08$ ; AI = 8.66 MRayl

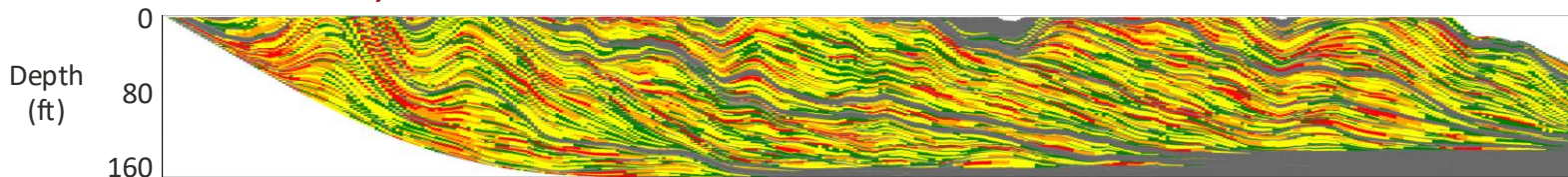


$\phi = 0.03$ ; AI = 8.69 MRayl

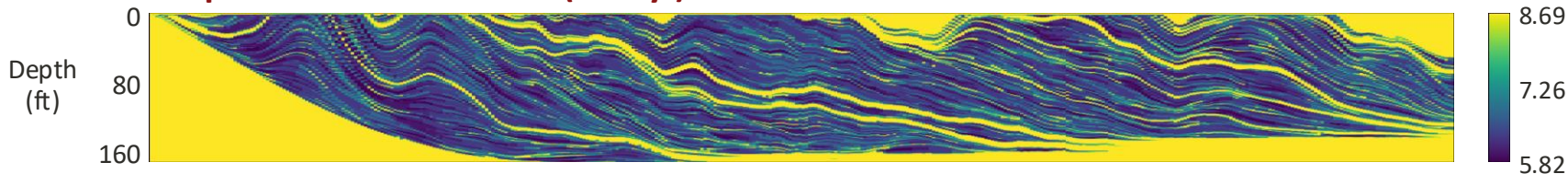
Porosity varies in all panels; clay content varies only in the first two.

# Facies, Impedance and Baseline Seismic

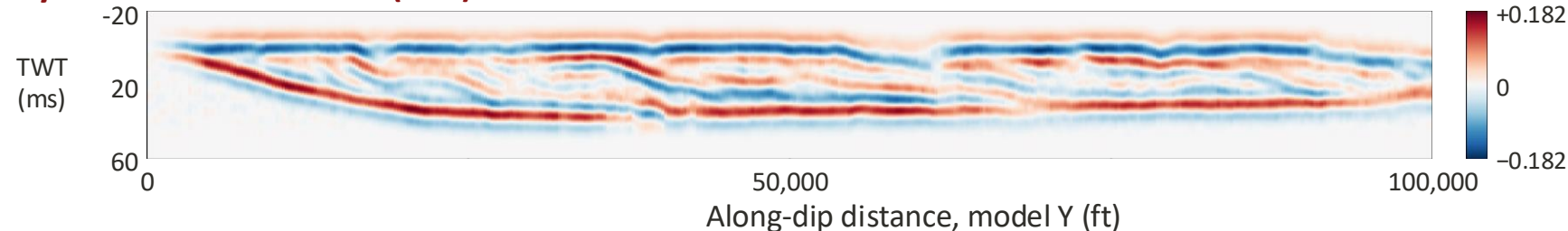
**Facies: native 0.04 ft; white = facies 99**



**Acoustic impedance: nominal 1 ft (MRayl)**

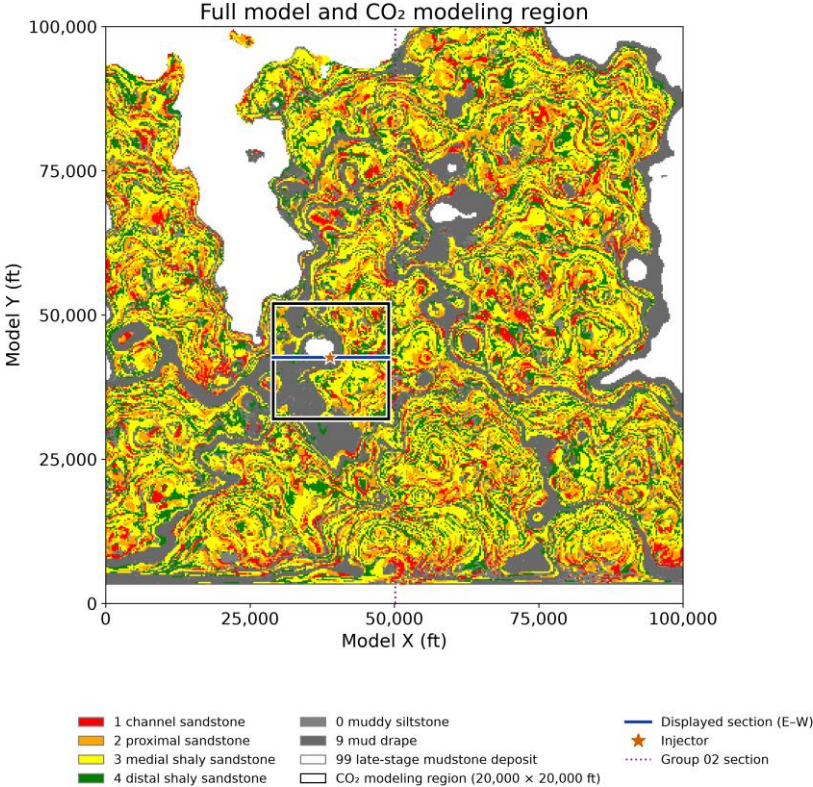


**Synthetic seismic: 1 ms (a.u.)**



Depth and model-top TWT are distinct coordinates. 60 Hz Ricker; independent vintage noise;  $2\lambda$  lateral Gaussian smoothing.

# CO<sub>2</sub> Forward-Modeling Case



| Setting          | Value                       |
|------------------|-----------------------------|
| Flow crop        | 80 × 80 × 1453 cells        |
| Extent           | 20,000 × 20,000 × 174.36 ft |
| Flow Δz          | 0.12 ft                     |
| Target injection | 0.5 Mt/year for 20 years    |
| Continuation     | 80 years after shut-in      |

**Same GAN realization; a larger crop than the resolution-study cases.**

Native geology Δz = 0.04 ft.

# Mineral and Fluid Reference Properties

Fixed reference properties used in forward modeling

| Material        | K (GPa) | G (GPa) | $\rho$ (g/cm <sup>3</sup> ) |
|-----------------|---------|---------|-----------------------------|
| Quartz          | 37      | 44      | 2.65                        |
| Equivalent clay | 21      | 7       | 2.58                        |
| Brine           | 2.2     | —       | 1.03                        |
| CO <sub>2</sub> | 0.1     | —       | 0.70                        |

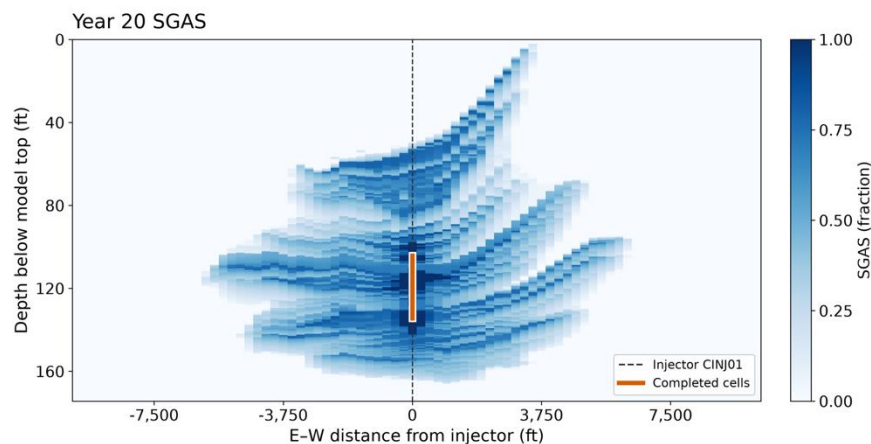
**Fluid mixing**

$$\frac{1}{K_f} = \frac{1 - SGAS}{2.2} + \frac{SGAS}{0.1}$$

$$\rho_f = (1 - SGAS) \times 1.03 + SGAS \times 0.70$$

flow pressure does not update the elastic frame.

# CO<sub>2</sub> Saturation and Fluid Substitution



## SGAS

Simulated gas saturation



## Fluid mixture

Wood Kf; saturation-weighted density



## Saturated elastic properties

Gassmann; fixed dry frame



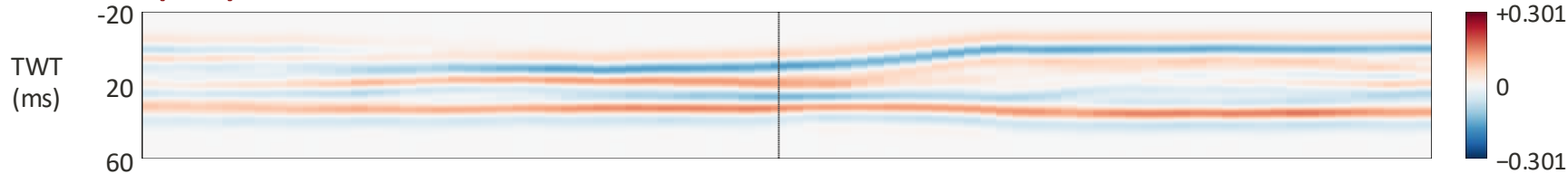
## Seismic response

Updated AI, travel times and reflectivity

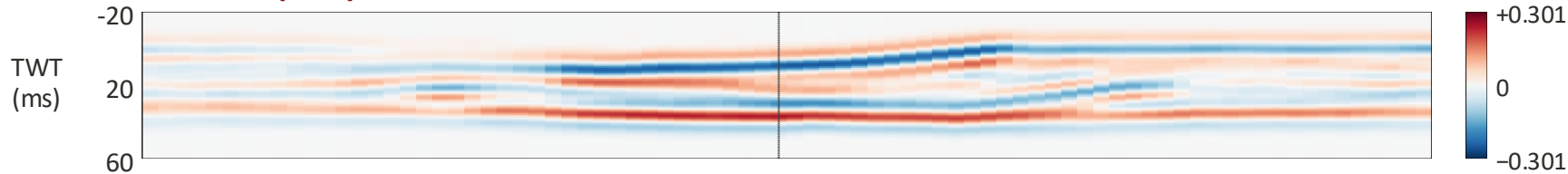
Porosity, mineral fractions and dry frame stay fixed; SGAS drives the elastic update.

# Year 20 Time-Lapse Seismic Response

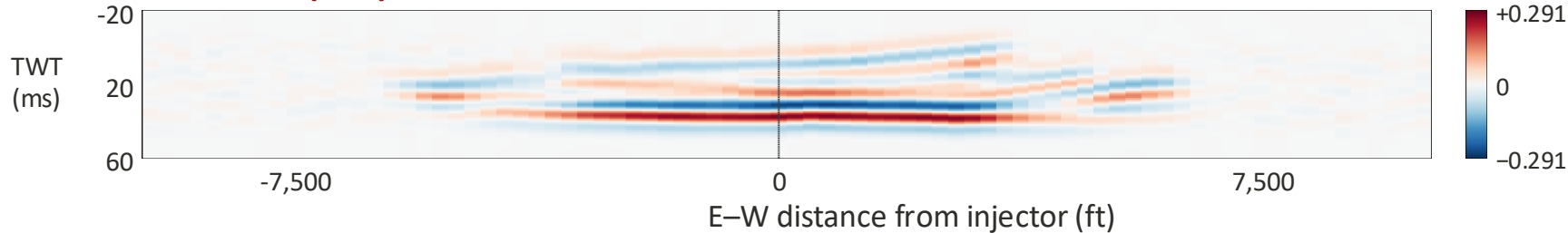
**Baseline (a.u.)**



**Year 20 monitor (a.u.)**



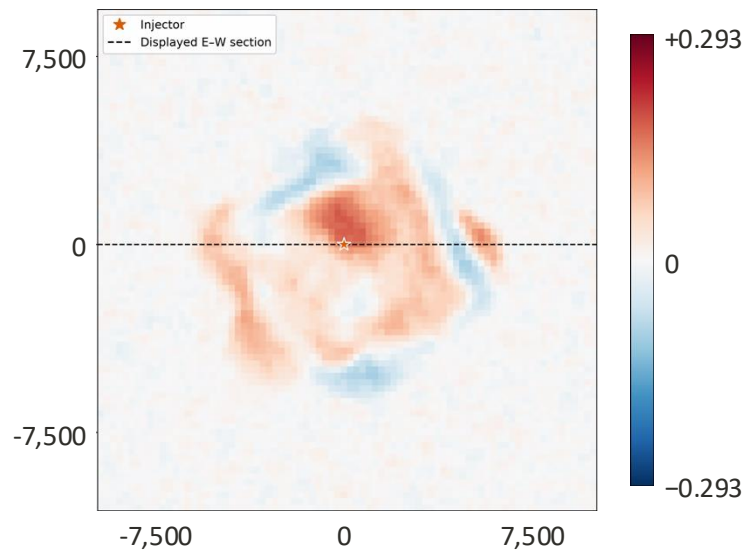
**Year 20 – Baseline (a.u.)**



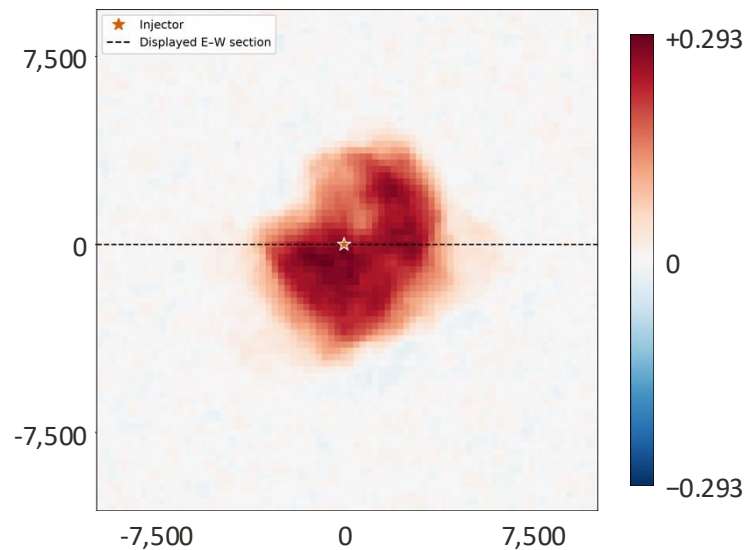
60 Hz Ricker; independent vintage noise;  $2\lambda$  lateral Gaussian smoothing. Baseline and monitor share a scale.

# Spatial Time-Lapse Seismic Response

**TWT = 25 ms**



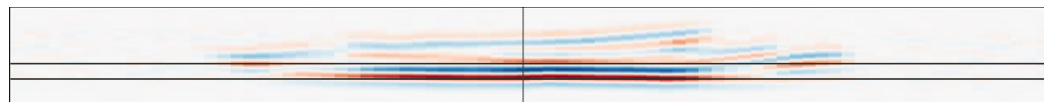
**TWT = 38 ms**



Year 20 – Baseline (a.u.); X/Y offsets from injector in ft

**Slice positions**

25 ms



38 ms



# Long-Term CO<sub>2</sub> and Seismic Evolution

**Year 20**

End of injection

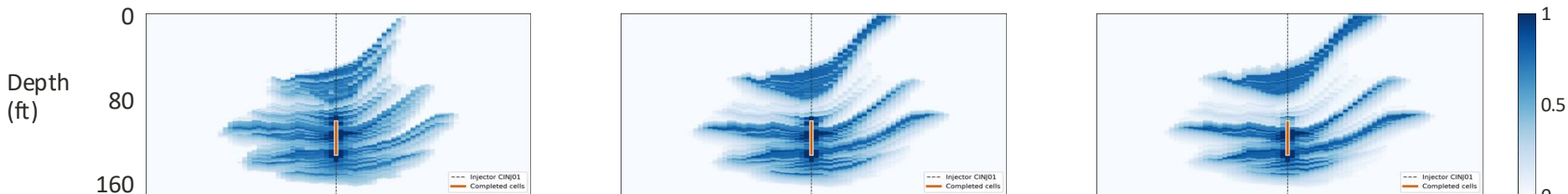
**Year 60**

40 years after shut-in

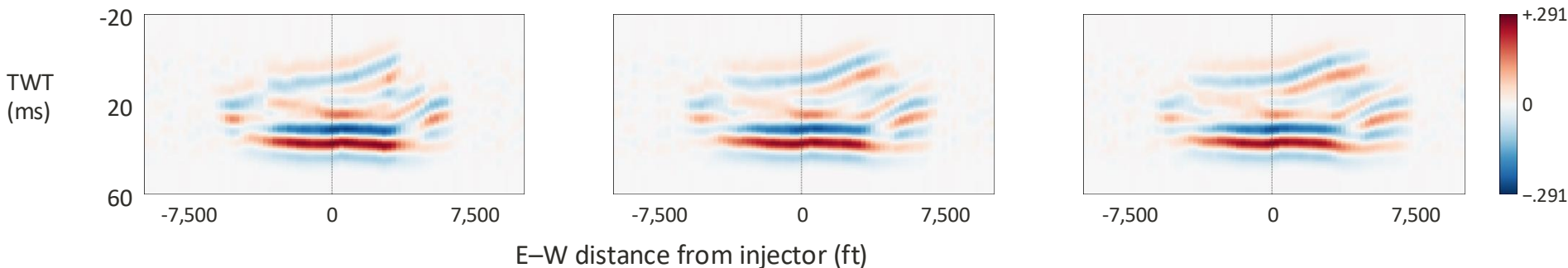
**Year 100**

80 years after shut-in

**SGAS (fraction)**



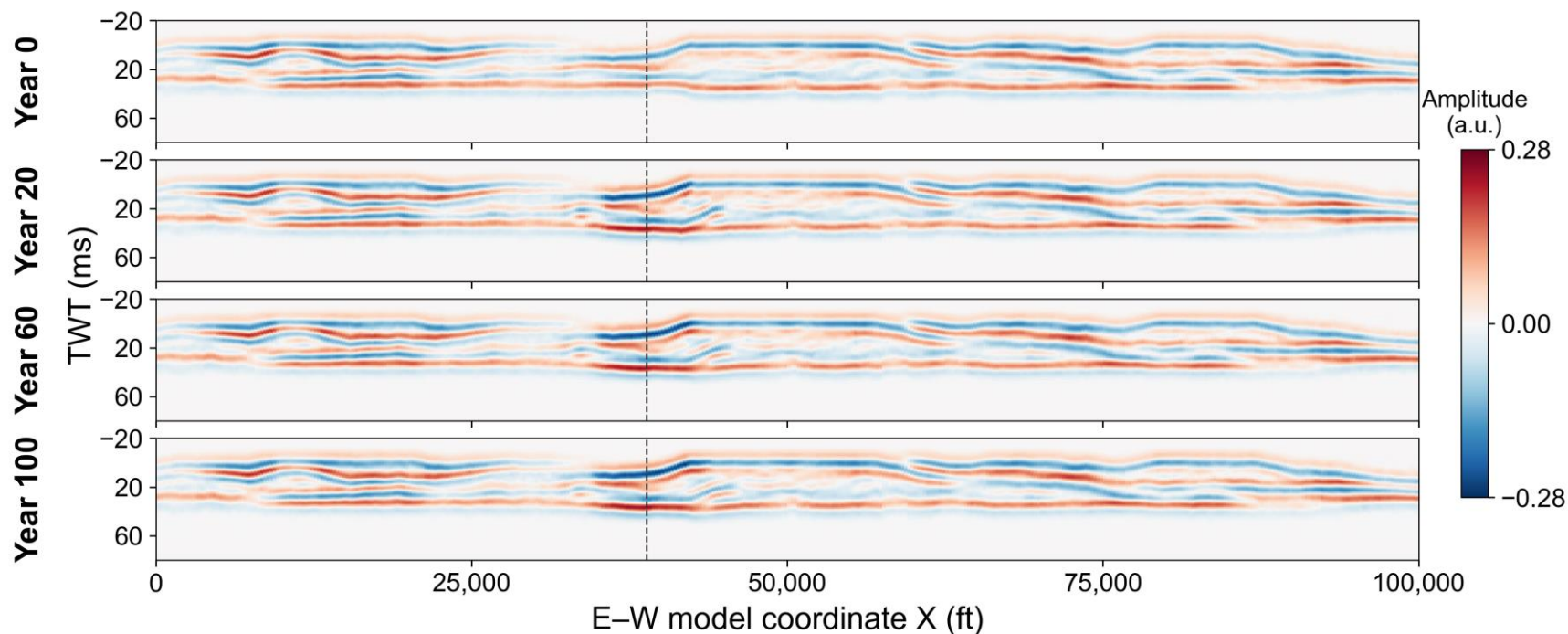
**Seismic difference from Year 0 (a.u.)**



Shared scales. Plume reaches the model top at Years 60 and 100; independent vintage noise is retained.

# Full-Model Seismic Sections

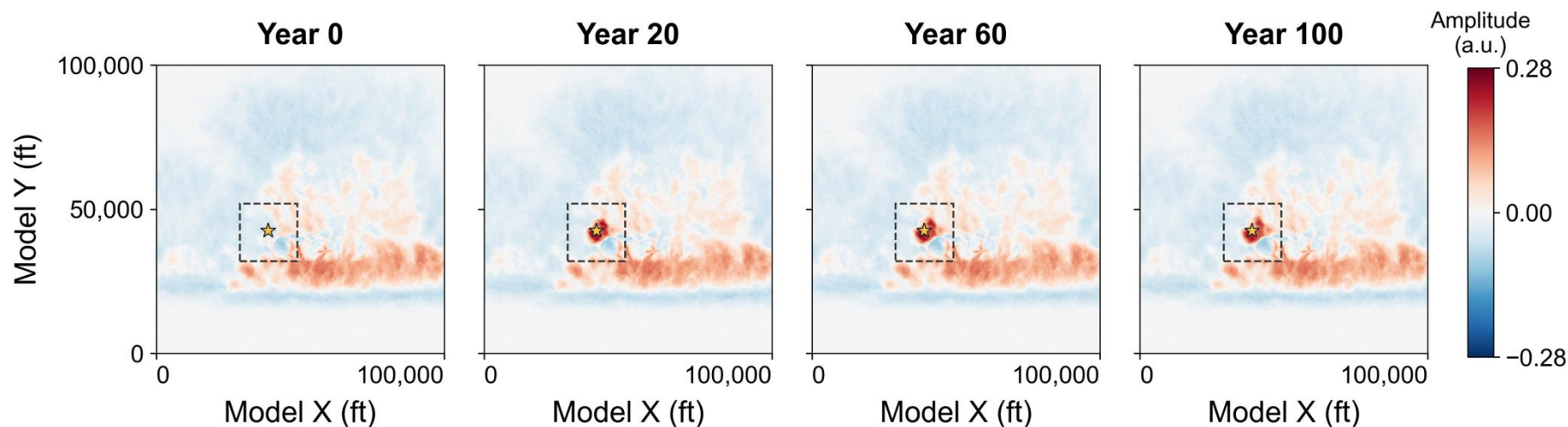
Year 0 baseline and Years 20, 60, 100. E–W through the injector.



60 Hz Ricker, 15% RMS noise and  $2\lambda$  lateral smoothing. Shared amplitude scale.

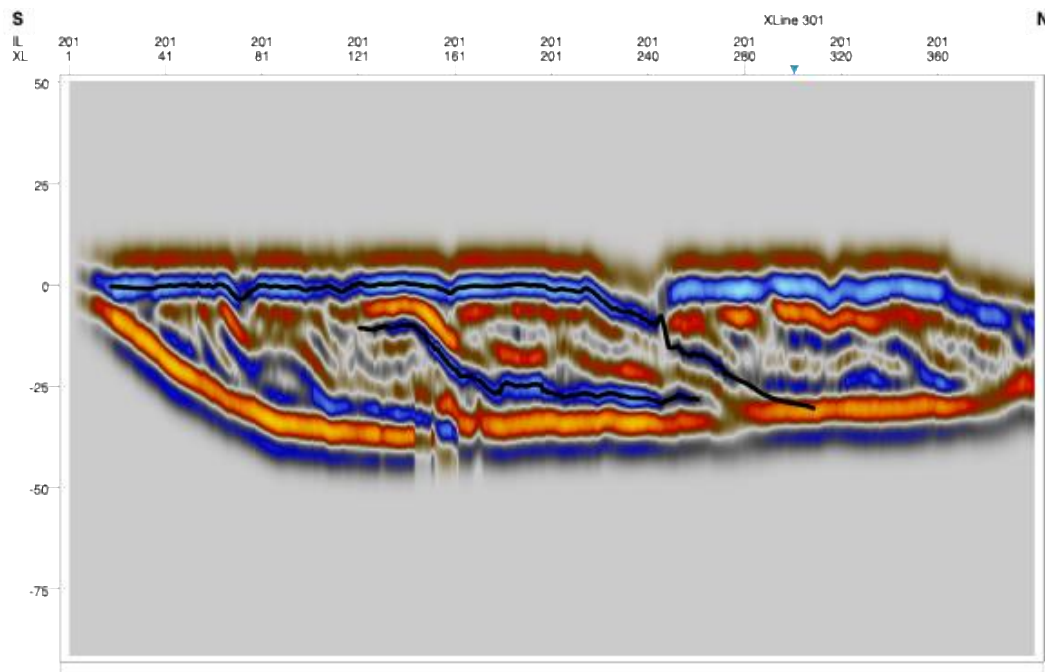
# Full-Model Seismic Time Slices

Year 0 baseline and Years 20, 60, 100. Full model at TWT = 38 ms.



Dashed box: flow-simulation crop. Star: injector. Shared amplitude scale.

# Horizon Interpretation in Petrel



## Synthetic seismic volume

$400 \times 400 \times 142$

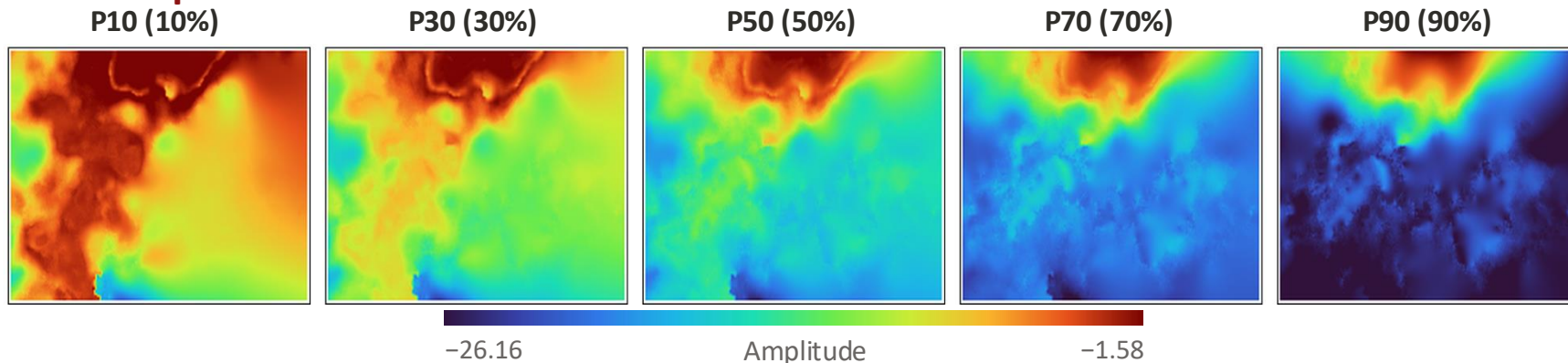
Inline / crossline  
1–400

Time sampling  
1 ms

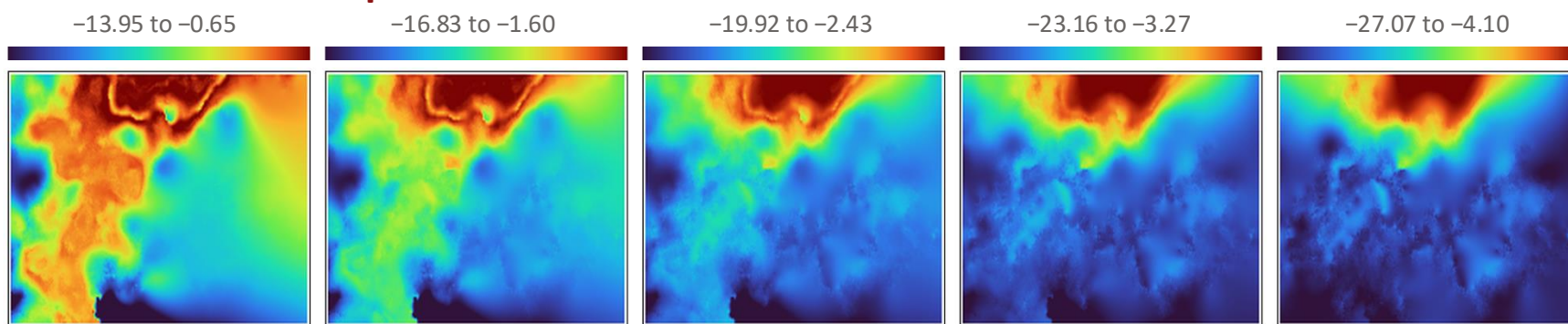
Interpreted horizons define surfaces for extracting stratigraphic amplitude slices.

# Horizon-Relative Amplitude Slices

## Shared amplitude scale



## Individual scales for spatial detail



# AI Agents

AI systems that use tools to carry out multi-step tasks.

Read the task and available context.

Choose actions and use tools.

Inspect the results and revise the next step.

## Examples

Codex, Claude Code, Antigravity

# Toward a Rock Physics Agent

## The Delta workflow motivates automation

| Stage              | Scientific decisions                   |
|--------------------|----------------------------------------|
| Rock physics       | Model assumptions                      |
| Fluid substitution | Saturation, fluid properties and units |
| Seismic modeling   | Sampling, wavelet and response checks  |

### Coding agents

Existing coding agents can execute many steps in this workflow.

### Experiments to guide development

Evaluate agents' ability to handle **complex rock physics models.**

# SRB MCP Architecture

## Coding agent

Plan the workflow

Prepare inputs

Interpret results

Requests and results



**MCP**

Model Context Protocol

## SRB rock physics service

Find models

Describe inputs

Run physics

The agent checks assumptions and interprets the returned physics outputs.



# Rock Physics Model Library

**74 operations across 19 model families**

## **Granular frames**

Hertz–Mindlin, soft sand

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## **Dispersion and attenuation**

Biot, White

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## **Effective medium models**

VRH, Hashin–Shtrikman

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## **Fluid properties / substitution**

Batzle–Wang, Gassmann

---

## **Stress and anisotropy**

Stress-dependent elasticity

---

## **Layering and reflection**

Backus, Zoeppritz

---

Selected capabilities. SRB 0.20.1 / interface 2.1.0, September 2026 snapshot.

---

# Model Knowledge and Validation

## Sources and assumptions

21 source-linked knowledge cards

---

## Parameters and units

Explicit input and output definitions

---

## Model-specific checks

Domain limits and numerical errors

---

## Execution records

Requests, inputs and traceability

---

The experiments assess how these resources support reliable rock physics workflows.

---

# A Model Call Through MCP

## 1 Search

Find a suitable model

srb\_search

---

## 2 Describe

Read units and assumptions

srb\_describe

---

## 3 Run

Execute with checked inputs

srb\_run

---

**Agent: interpret outputs and verify derived quantities**

# Complex Rock Physics Models: Agent Evaluation

45 independent runs; three repetitions per resource condition

| Case         | GPT-5.6 Sol / high    | GPT-6 Sol / high      | Runs |
|--------------|-----------------------|-----------------------|------|
| Smeaheia     | 3 conditions × 3 runs | Not run               | 9    |
| Glass beads  | 3 conditions × 3 runs | 3 conditions × 3 runs | 18   |
| White patchy | 3 conditions × 3 runs | 3 conditions × 3 runs | 18   |

| Resource condition   | Reference textbook | SRB via MCP |
|----------------------|--------------------|-------------|
| No textbook / no MCP | No                 | No          |
| Textbook             | Yes                | No          |
| Textbook + MCP       | Yes                | Yes         |

All groups receive the task, public inputs, output requirements and ordinary coding tools.

# Smeaheia: Inputs and Task

## Oil–brine substitution in a conditional synthetic benchmark

### Input data

Three calibration wells with elastic logs and synthetic seismic.  
BLIND\_D: prediction targets hidden.

All wells: depth, TWT, porosity, shale fraction and water saturation at baseline and monitor states.

Mineral/fluid properties and state.  
20 Hz Ricker, 2 ms, angles 1–45°.

### Task for the agent

Select appropriate physical models.  
Predict BLIND\_D elastic properties, PP reflectivity and synthetic seismic at baseline and monitor states.

Report the seismic difference and explain the saturation response.

450 elastic rows, 20,160 reflections.  
Three 1000 × 45 seismic gathers.

Tests model selection and consistency across a complete forward-modeling chain.

# Smeaheia: Physical Model

## Material assumptions

Uncemented, rough-contact grains.  
Porosity and pressure control  
frame stiffness.

## Fluid substitution

Connected pores, equilibrated pressure.  
Substitution changes  $K_{sat}$  and density.  
Gassmann keeps  $G_{sat} = G_{dry}$ .

## Seismic response

Elastic contrasts produce angle-dependent  
PP reflections, then wavelet convolution.

## Reference modeling chain

### VRH

Effective mineral properties

### Hertz–Mindlin

Granular dry-frame stiffness

### Batzle–Wang + Wood

Fluid properties and mixing

### Gassmann

Saturated elastic properties

### Full Zoeppritz PP

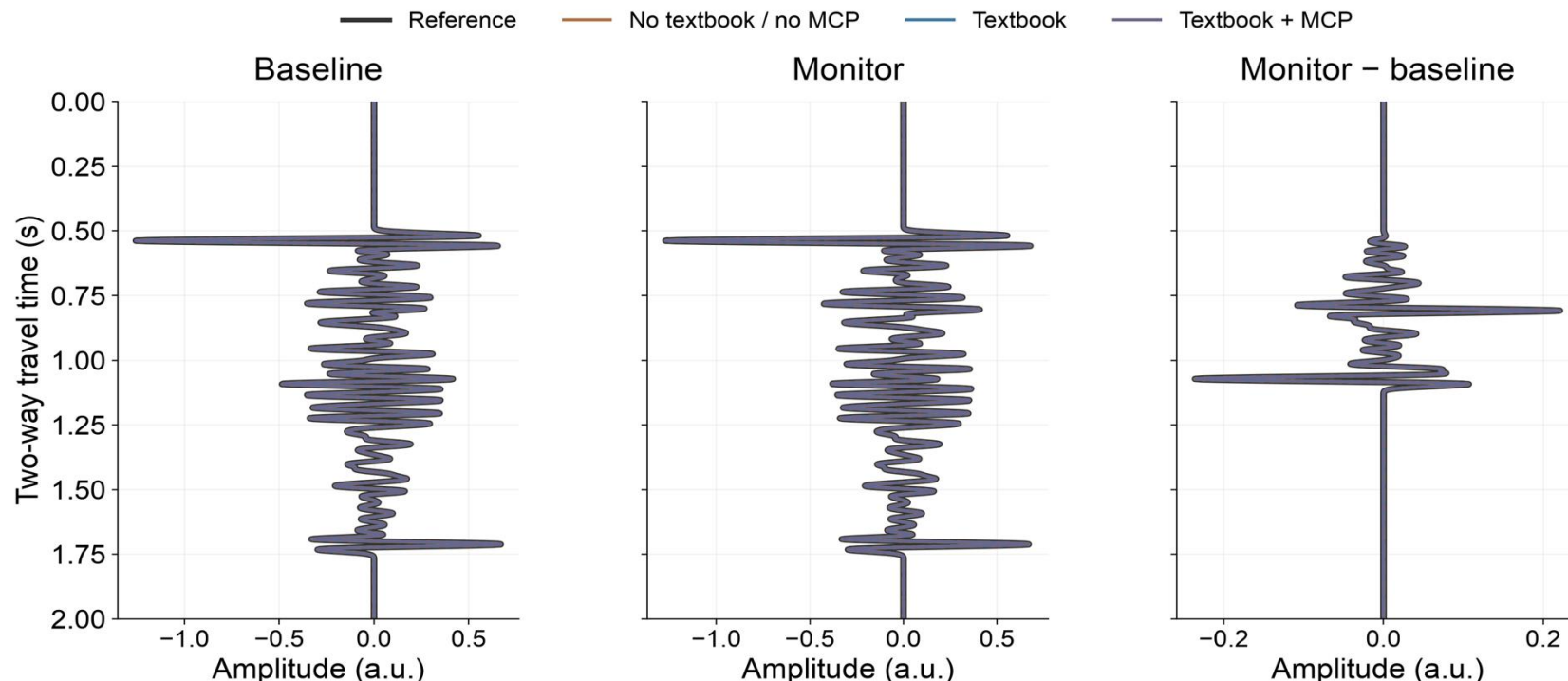
Angle-dependent reflectivity

### Interpolation + convolution

Synthetic seismic gathers

# Smeaheia: Complete Workflow Agreement

9/9 runs reproduce the numerical reference



BLIND\_D, 30° PP. Separate amplitude scales. Maximum seismic NRMSE  $< 1.68 \times 10^{-10}$ .

# Glass Beads: Inputs and Task

**Fully water-saturated, isotropic porous glass-bead sample**

## Given nominal inputs

| Input     | Values                                                                       |
|-----------|------------------------------------------------------------------------------|
| Mineral   | $K = 43.7 \text{ GPa}$ ; $\rho = 2445 \text{ kg/m}^3$                        |
| Dry frame | $K = 4.826$ ; $G = 3.614 \text{ GPa}$                                        |
| Pores     | $\phi = 0.391$ ; tortuosity = 1.44                                           |
| Flow      | $\kappa = 19.9 \times 10^{-12} \text{ m}^2$ ; $r_0 = 23 \text{ }\mu\text{m}$ |
| Water     | $K = 2.209 \text{ GPa}$ ; $\rho = 995 \text{ kg/m}^3$                        |
| Viscosity | $\eta = 0.001 \text{ Pa s}$                                                  |

## Task for the agent

Choose a model and predict fast P, slow P and S across 1 Hz–1 MHz.

For each mode: phase velocity, inverse Q and spatial attenuation.

Also resolve 0.5–1 MHz and report the nine predictions at 780 kHz.

**Observed responses are withheld.  
No fitting or input adjustment.**

Tests frequency-dependent physics, mode identification and output definitions.



# Biot: Coupled Solid and Fluid Motion

**A deformable porous frame and its pore fluid form two coupled continua.**

**Frame displacement:  $u$**

Relative fluid displacement:  $w = \varphi(U - u)$

$U$ : fluid displacement. Stress and pore pressure couple the motions; inertia and viscous drag determine their frequency response.

| Mode   | Typical motion / behavior                                 |
|--------|-----------------------------------------------------------|
| Fast P | Compression; frame and fluid move largely together        |
| Slow P | Strong relative motion; diffusive at low frequency        |
| S      | Transverse motion; fluid inertia and drag also contribute |

Linear, isotropic poroelasticity. Biot (1956); Johnson et al. (1987).

# Biot: Cylindrical Pores and JKD

Same Biot coupling; different frequency-dependent viscous resistance.

$$b(\omega) = \frac{\eta}{\kappa} F(\omega)$$

Low-frequency Darcy limit:  $F \rightarrow 1$

$\eta$ : fluid viscosity;  $\kappa$ : static permeability;  $F$ : viscous correction.

|              | Cylindrical pores                           | Biot–JKD                                           |
|--------------|---------------------------------------------|----------------------------------------------------|
| Geometry     | Identical circular tubes                    | Effective pore geometry                            |
| Flow model   | Exact tube-flow solution (Bessel functions) | Square-root approximation matching low/high limits |
| Length input | Tube radius $R$                             | Viscous length $\Lambda$                           |

For  $\Lambda = R$ , JKD still differs from the exact Bessel response.

Biot (1956); Johnson et al. (1987); Chandrasekaran et al. (2022).

# Glass Beads: Model and Length Choices

No textbook / no MCP: three independent runs per model

| Model   | R1                   | R2                   | R3                    |
|---------|----------------------|----------------------|-----------------------|
| GPT-5.6 | JKD, $\Lambda = r_o$ | JKD, $\Lambda = r_o$ | JKD, $\Lambda = 2r_o$ |
| GPT-6   | Bessel               | JKD, $\Lambda = r_o$ | JKD, $\Lambda = 2r_o$ |

Both R3 reports call  $\Lambda = 2r_o$  a cylindrical-pore proxy.

Textbook only: all six runs select the cylindrical-pore reference.

Does the cylindrical-pore assumption  
justify  $\Lambda = 2r_o$ ?

Model selection and parameter mapping are separate judgments.

# Viscous Length: Radius or Diameter?

**Definition: pore volume / wetted wall area, weighted by inviscid flow.**

$$\Lambda = 2 \frac{\int_V |v_\infty|^2 dV}{\int_{S_w} |v_\infty|^2 dS}$$

$v_\infty$ : inviscid pore velocity;  $S_w$ : wetted wall, excluding open ends.

**Uniform straight circular tube of radius  $r_0$  and length  $L$**

$$\Lambda = \frac{2V}{S_w} = \frac{2\pi r_0^2 L}{2\pi r_0 L} = r_0$$

**The factor 2 is already included. The result is the radius.**

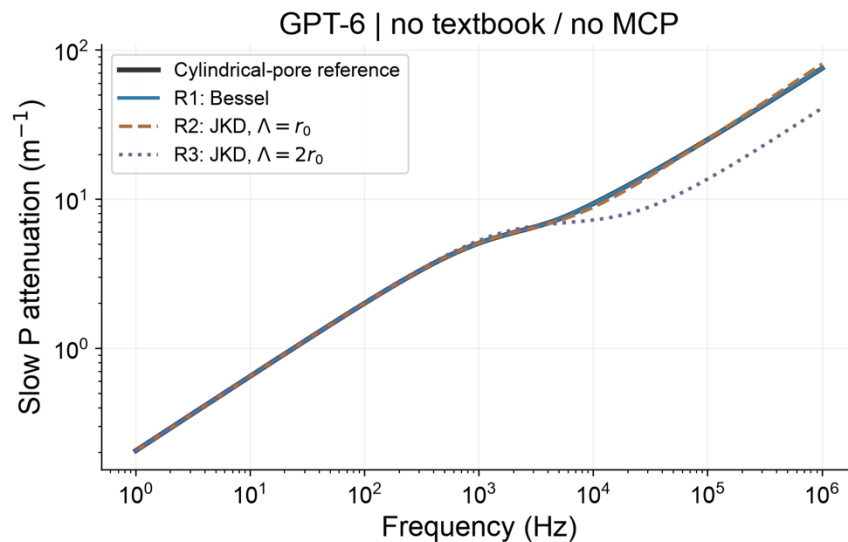
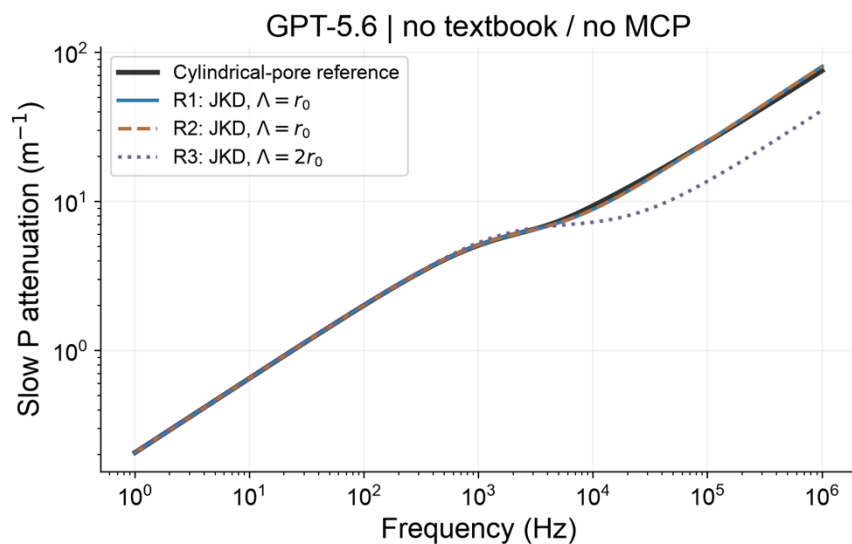
A measured pore radius need not equal  $\Lambda$  in a real pore network.

$\Lambda = 2r_0$  requires additional evidence beyond a cylindrical-pore label.

Johnson et al. (1986, 1987); input  $r_0 = 23 \mu\text{m}$  is a radius.

# Glass Beads: Variation Across Runs

## No textbook / no MCP: slow-P attenuation



Different pore-scale assumptions change the predicted response.

Differences include model choice and the  $\Lambda$  mapping; they do not establish a model ranking.

# Glass Beads: Textbook-Based Method

**Textbook only: consistent model and output definitions in all six runs**

## **1 Use the cylindrical-pore Biot model**

Bessel-function viscous drag. Use the supplied  $r_0$  as the effective tube radius.

## **2 Solve complex wave numbers for fast P, slow P and S**

$$k = k_r - i\alpha, \quad v = \frac{\omega}{k_r}, \quad \alpha = -\text{Im}(k)$$

## **3 Define inverse Q from the same wave number**

$$q_k = \frac{2\alpha}{k_r} \Rightarrow \alpha = \frac{\pi f q_k}{v}$$

$\alpha$  measures spatial amplitude decay ( $\text{m}^{-1}$ ). Handbook §6.1 and six textbook-only submissions.

# Glass Beads: Six MCP Runs

**Textbook + MCP: the inverse-Q interpretation error occurred in 2 of 6 runs**

| Model   | R1    | R2      | R3      |
|---------|-------|---------|---------|
| GPT-5.6 | Error | Correct | Correct |
| GPT-6   | Error | Correct | Correct |

**Error: tool inverse Q treated as wave-number inverse Q.**

Correct: exact nonlinear conversion of the tool’s inverse Q.

All six selected `biot_full_frequency`. The difference arose after the tool call.

Status refers to Q-to-attenuation conversion. R1–R3 are independent runs, not sequential learning.

# Where the Q Definition Changes

## Same field name, different mathematical definitions

### Returned by the frozen MCP model

$$q_T = \frac{|\text{Im}(c^2)|}{\text{Re}(c^2)}, \quad c = \frac{\omega}{k}$$

Loss tangent of complex squared velocity

### Assumed by the two R1 agents

$$q_T = q_k = \frac{2\alpha}{k_r}$$

Wave-number ratio, defined from spatial decay

**The mismatch enters when the returned  $q$  is inserted here:**

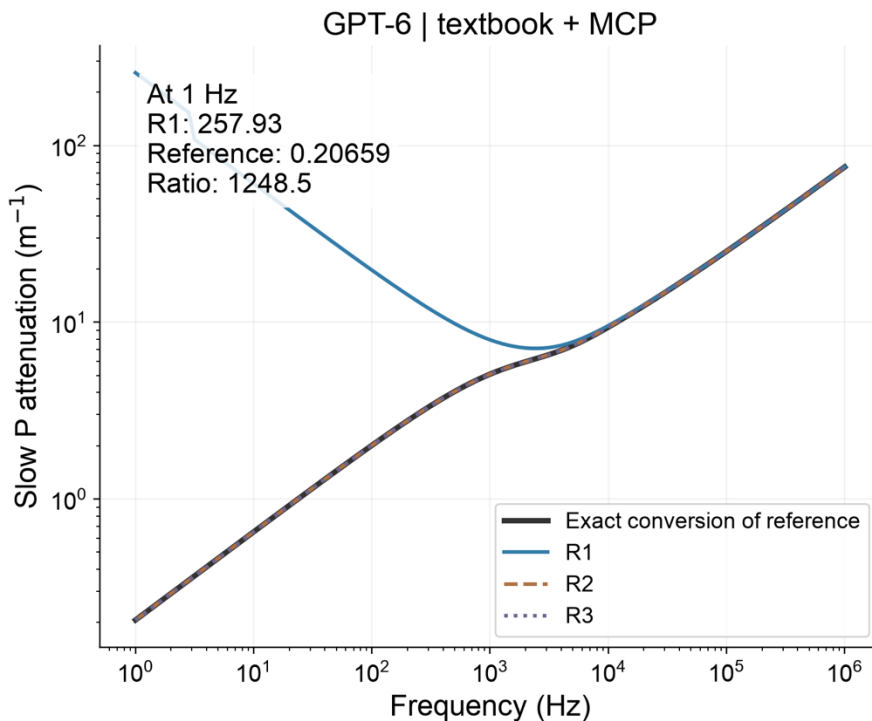
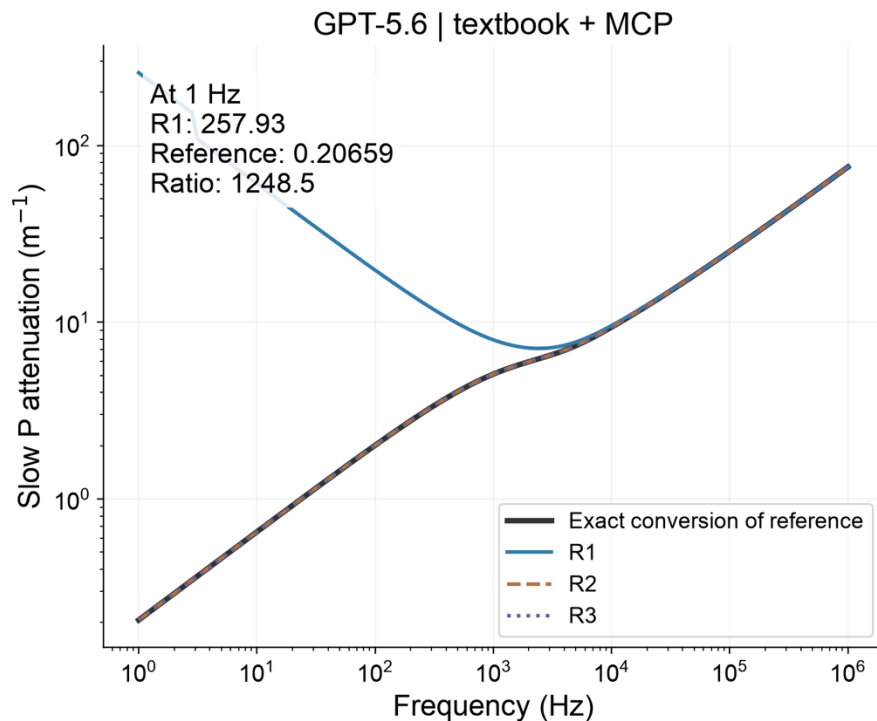
$$\alpha_{\text{used}} = \frac{\pi f q_T}{v}$$

Frozen biot\_full\_frequency source and both R1 scripts. The output label did not spell out the algebra.



# Effect of Misinterpreting Inverse Q

## Two R1 runs overestimate low-frequency slow-P attenuation



Correct conversions in R2 and R3 recover the reference response.

# White: Inputs and Task

## Six idealized CO<sub>2</sub>–water cases with fixed material properties

### Input data

Frame/mineral moduli and density,  
porosity and permeability.  
CO<sub>2</sub>/water moduli, density and viscosity.

### Task for the agent

Implement White pressure diffusion  
with the Dutta–Seriff correction.  
Predict complex K, V<sub>p</sub>, attenuation  
and low/high-frequency K limits.

| CO <sub>2</sub> saturation | Radius 1.25 cm | Radius 2.5 cm | Radius 5 cm |
|----------------------------|----------------|---------------|-------------|
| 0.1%                       | W11            | W12           | W13         |
| 0.5%                       | W21            | W22           | W23         |

Radius = inner CO<sub>2</sub> sphere. 121 frequencies per case, 726 targets in total.

The model is prescribed. Test faithful implementation, without fitting or added loss.

# White: A Spherical Patch Model

**An effective model for rock containing separate gas- and water-saturated patches.**

| Region                        | Saturation and geometry                            |
|-------------------------------|----------------------------------------------------|
| Inner sphere, radius $a$      | Same porous frame; pores filled with $\text{CO}_2$ |
| Outer shell, outer radius $b$ | Same porous frame; pores filled with water         |

Equal porosity in both regions:  $S_g = \left(\frac{a}{b}\right)^3$

**Compression  $\rightarrow$  pressure contrast  $\rightarrow$  Darcy flow between patches**

Volume-averaged response:  $K^*(\omega) = K'(\omega) + iK''(\omega)$

$K'$ : stiffness;  $K''$ : loss. Real shear modulus stays fixed.

Finite fluid moduli; closed outer boundary; quasi-static, long-wavelength.

White (1975), with Dutta–Seriff correction (1979); task-specified geometry.

# White: Misinterpreting the Specified Model

The task specified corrected White pressure diffusion.

The agent treated a quasi-static Biot solution as that model.

**Pressure diffusivity:** 
$$D_j = \frac{\kappa}{\eta_j C_j}$$

**Required: corrected White**

$$C_j = \frac{1}{M_j} + \frac{\alpha_B^2}{K_d}$$

Bulk frame modulus  $K_d$

**Agent: quasi-static Biot BVP**

$$C_j = \frac{1}{M_j} + \frac{\alpha_B^2}{K_d + \frac{4}{3} G_d}$$

Longitudinal frame modulus

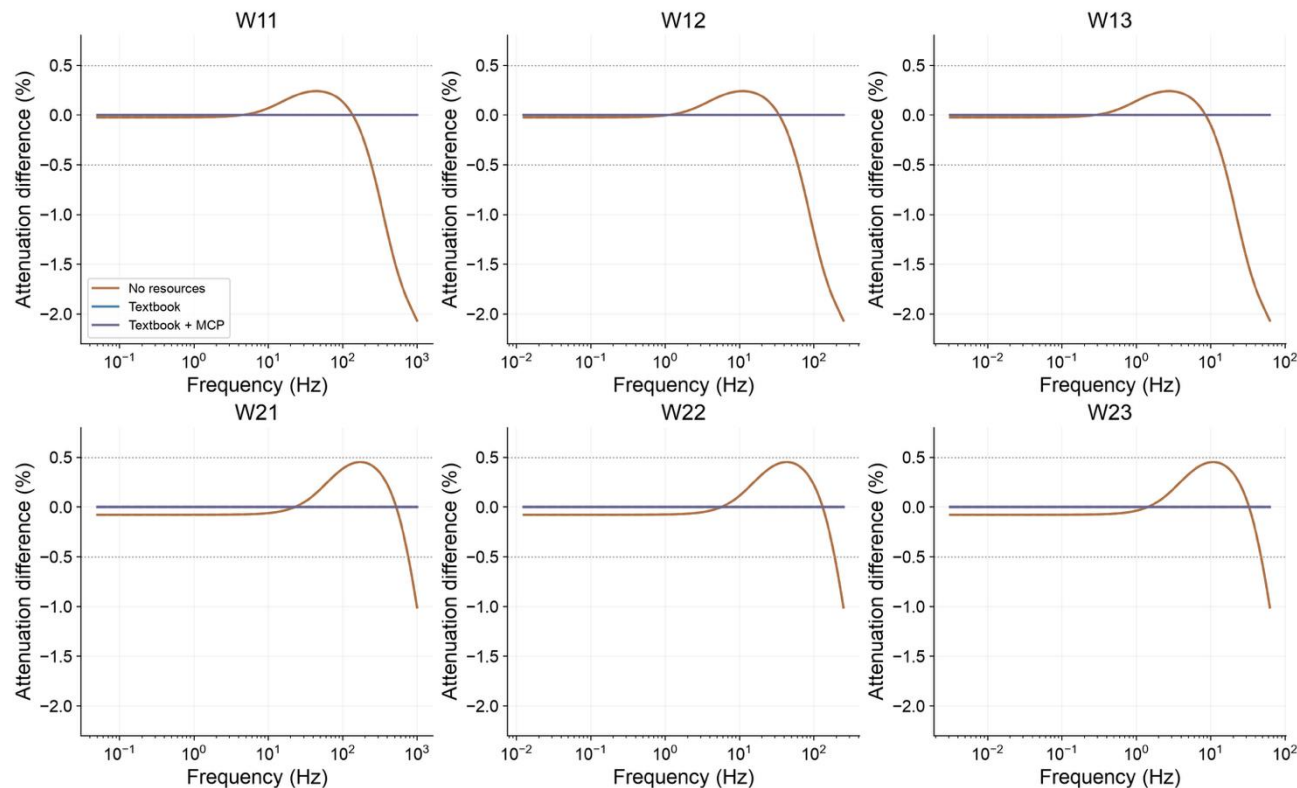
$C_j$ : storage;  $\alpha_B$ ,  $M_j$ : Biot coefficient, modulus;  $K_d$ ,  $G_d$ : dry bulk, shear moduli.

**A different storage term changes the model being solved.**

**Correct limits and numerical checks do not establish task compliance.**

Task specification and submitted report/code; Vogelaar et al. (2010).

# White: A Systematic Model Mismatch



No textbook / no MCP

Misidentified the specified model

Limits agree, but the equations differ

**2.07%**

maximum attenuation error vs target model

Textbook and MCP:  
Reproduce the required model

Solid: GPT-6; dashed: GPT-5.6. All three runs.  $\pm 0.5\%$  lines show only the relative tolerance.

# What the Agent Experiments Show

## No reference resources

Glass beads: an unsupported mapping from pore radius to viscous length.  
White: a related Biot formulation mistaken for the prescribed model.

## Textbook guidance

Matches the reference calculations achieved with correctly used MCP.  
Helps identify and implement complex rock physics models in these tests.

## Textbook + MCP

Tools still require correct interpretation.  
Glass beads: inverse-Q conversion failed in both R1s.

**Smeaheia: all nine runs succeeded. Next: textbook + coding agent on Delta.**

# Delta Agent: Inputs and Task

**GPT-5.6 Sol / high with the Handbook, without MCP**

## **Input data**

GAN facies volume:  $400 \times 400 \times 4359$ .  
ECLIPSE CO<sub>2</sub> states on the flow crop.  
Ten vintages from Year 0 to Year 100.

## **Fixed processing requirements**

60 Hz Ricker and 15% RMS noise.  
 $2\lambda$  lateral Gaussian smoothing.  
Large jobs through SLURM.

## **Task for the agent**

Select and document the rock physics.  
Map facies and fluid states to Vp, Vs,  
density, impedance and reflectivity.

Generate full seismic and differences  
relative to Year 0 for every vintage.  
Deliver code, arrays, figures and checks.

**The processing chain is given.**  
**Physical choices and coding are open.**

Previous scripts are plotting references only. This is a separate application test.

# Delta: Physical Assumptions and Choices

## **Facies need physical properties**

Facies labels describe geology. Elastic prediction also needs porosity, mineral fractions, grain-contact assumptions and effective pressure.

## **Fluid substitution has an assumption set**

In the adopted Wood–Gassmann workflow, mixed-fluid properties update saturated bulk modulus and density while the dry frame stays fixed.

## **Open choices affect the entire seismic volume**

The agent selects frame parameters, fluid treatment and depth discretization. These choices affect impedance contrasts and travel times before injection.

Compare the completed workflow and its assumptions with our modeling choices next.



# Delta: Modeling Choices and Their Effects

Shared physics: VRH, Hertz–Mindlin / modified HS, and Gassmann

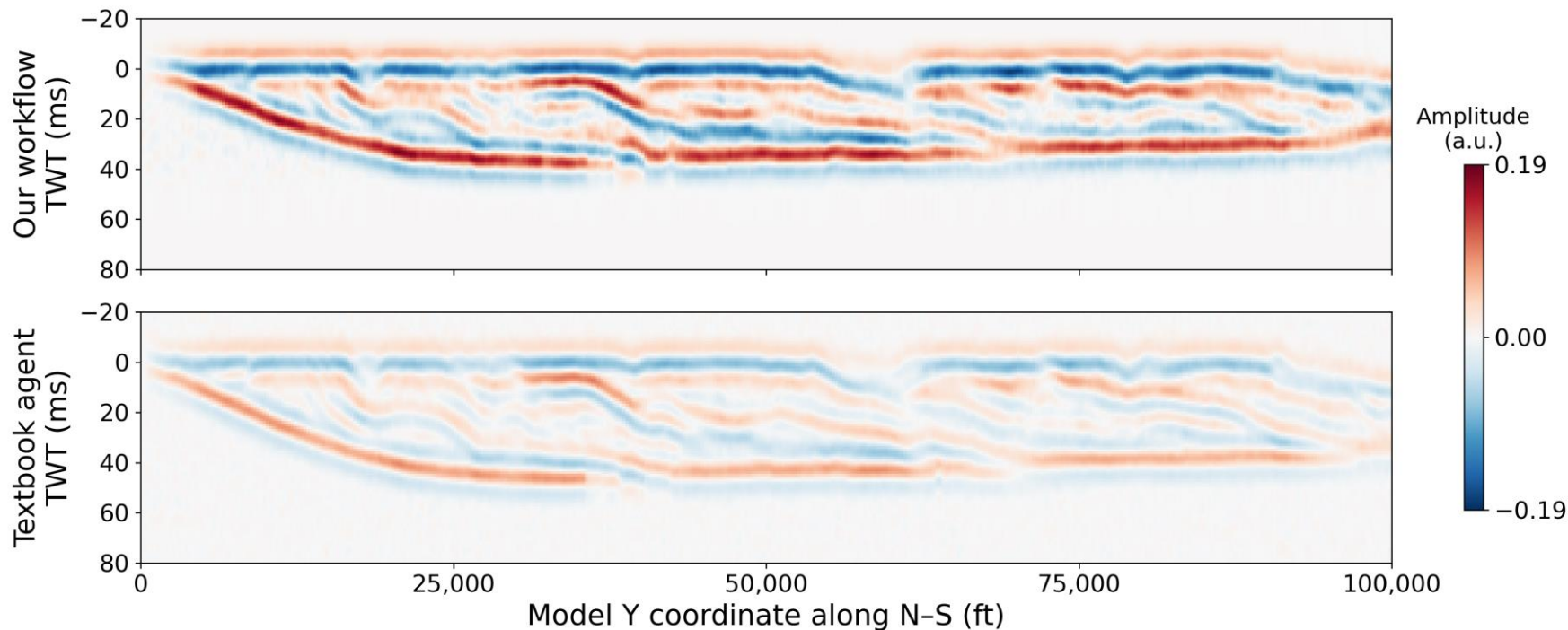
| Modeling choice    | Our workflow                             | Textbook agent                           |
|--------------------|------------------------------------------|------------------------------------------|
| Frame structure    | Three facies-specific groups             | One frame parameter set                  |
| Porosity $\phi$    | Sand: 0.18–0.27<br>Silt / mud: 0.03–0.08 | Sand: 0.28–0.36<br>Silt / mud: 0.18–0.24 |
| Effective pressure | 20 MPa, fixed                            | 8 MPa, fixed                             |
| Prior selection    | Facies priors + clay scenarios           | One assumption set                       |
| Elastic depth grid | Upscaled to about 1 ft                   | Fine grid: 0.04 ft                       |

**Different assumptions change impedance and travel time, producing different baseline and monitor seismic.**

Textbook implementation alone does not resolve geological model selection.

# Year 0 Seismic Across the Full Model

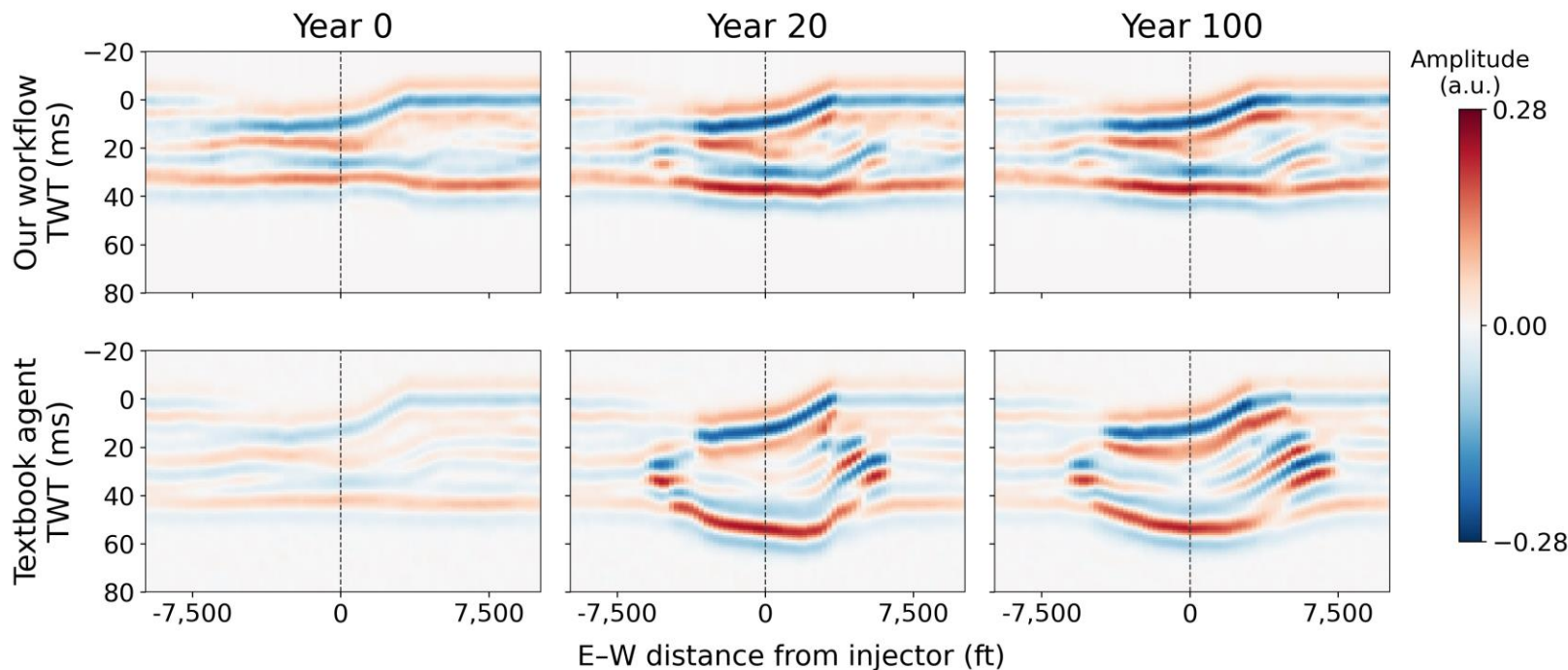
N–S section at X = 50,125 ft. Full model width: 100,000 ft.



**Background seismic differs across the model before CO<sub>2</sub> injection.**

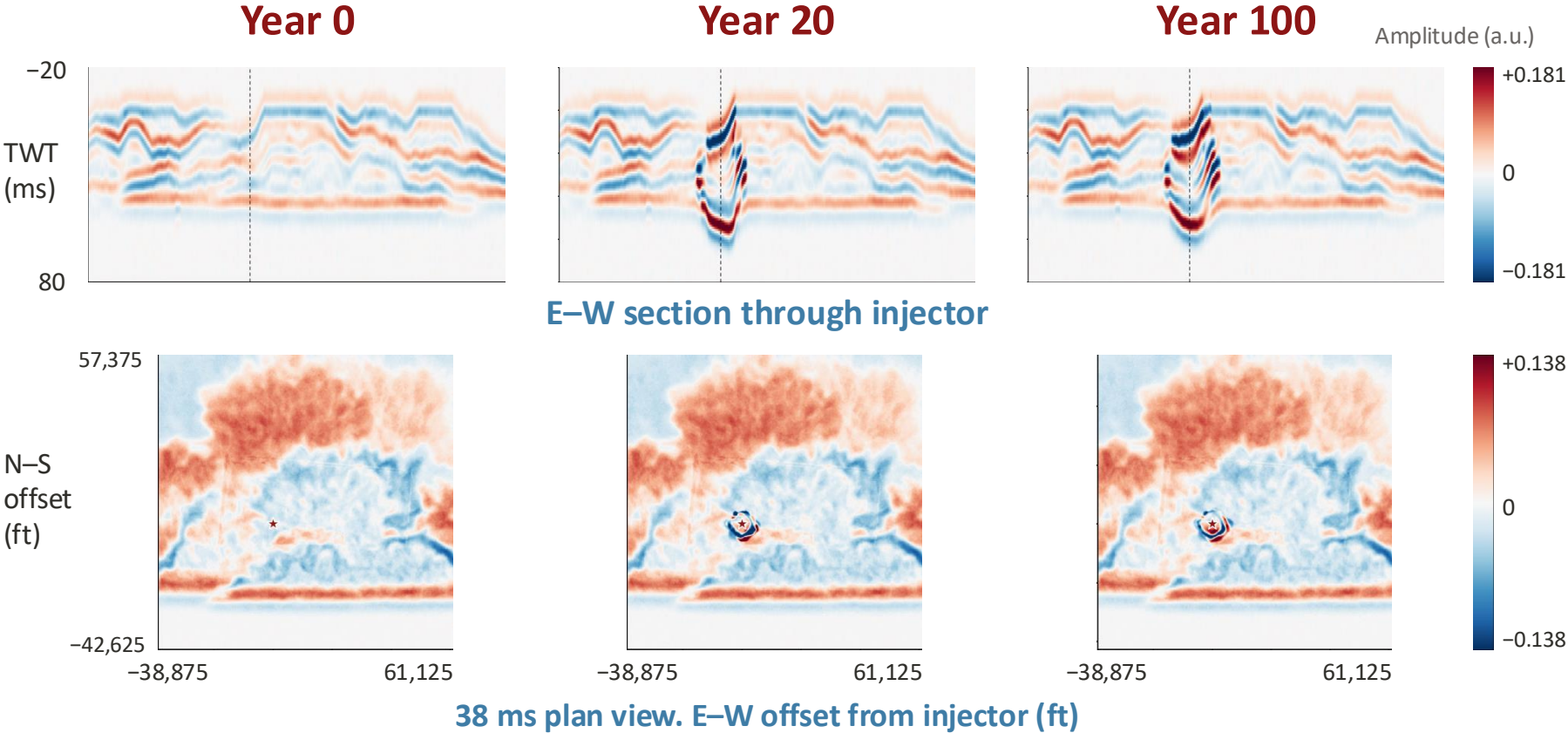
# Delta Seismic: Our Workflow and the Agent

E–W section through the injector. Full seismic, with one shared amplitude scale.



**Differences are already present at Year 0, before CO<sub>2</sub> injection.**

# Delta Agent: Seismic Results



# Next Step: Developing the Rock Physics Agent

**Build on the capabilities of existing coding agents**

## **Software-specific tools**

Add interfaces for specialist software and data exchange, including Petrel and ECLIPSE workflows.

## **Multi-agent collaboration**

Seismic agent, well-log agent... and coding agent.  
contribute domain knowledge and scientific checks.

A dedicated coding agent implements, runs and debugs the workflow.

# Future Research with Forward Modeling

## **Main research focus**

Continue developing and a rock physics agent.

## **Turbidite channel datasets**

Generate paired synthetic seismic and facies data through forward modeling.

## **Value of seismic information (VOI)**

Assess how seismic observations distinguish geological realizations and support decisions relative to acquisition cost.